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*(Information)*INFORMATION FROM EUROPEAN UNION INSTITUTIONS, BODIES, OFFICES
AND AGENCIES

EUROPEAN COMMISSION

EXPLANATORY NOTES TO THE COMBINED NOMENCLATURE OF THE
EUROPEAN UNION*(2019/C 0/01)*

**Publication made in accordance with Article 9(1) of Council Regulation (EEC) No 2658/87
of 23 July 1987 on the tariff and statistical nomenclature and on the Common Customs
Tariff⁽¹⁾**

⁽¹⁾ OJ L 256, 7.9.1987, p. 1.

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FOREWORD

Council Regulation (EEC) No 2658/87 of 23 July 1987 on the tariff and statistical nomenclature and on the Common Customs Tariff ⁽¹⁾ established a nomenclature, known as the 'Combined Nomenclature' or abbreviated to the 'CN', based on the International Convention on the Harmonised Commodity Description and Coding System ⁽²⁾, known as 'the Harmonised System' or abbreviated to the 'HS'.

The HS has been supplemented by its own Explanatory Notes (HSEs). Those notes in English and French are issued and kept up to date by the:

WORLD CUSTOMS ORGANISATION (WCO)

Customs Cooperation Council (CCC),

30, rue du Marché,

B-1210 Brussels

By virtue of Article 9(1)(a), first indent, of Regulation (EEC) No 2658/87, the Commission adopts explanatory notes to the Combined Nomenclature (CNENs), following consideration by the Tariff and Statistical Nomenclature Section of the Customs Code Committee. Although the CNENs may refer to the HSEs, they do not take the place of the latter, but should be regarded as complementary to and used in conjunction with them.

This edition of the CNENs includes and, where appropriate, replaces those published in the *Official Journal of the European Union*, C series, up to 4 January 2019 ⁽³⁾. CNENs published in the Official Journal, C series, subsequent to that date remain in force and will be incorporated in the CNENs when revised.

In addition, the CN heading and subheading codes to which reference is made reflect those codes in the Combined Nomenclature for 2019, as set out in Commission Implementing Regulation (EU) 2018/1602 ⁽⁴⁾.

Furthermore, information concerning 'Guidelines on the classification in the Combined Nomenclature of goods put up in sets for retail sale' was published in the *Official Journal of the European Union*, C series ⁽⁵⁾.

⁽¹⁾ OJ L 256, 7.9.1987, p. 1.

⁽²⁾ OJ L 198, 20.7.1987, p. 1.

⁽³⁾ OJ C 2, 4.1.2019, p. 2.

⁽⁴⁾ OJ L 273, 31.10.2018, p. 1.

⁽⁵⁾ OJ C 105, 11.4.2013, p. 1.

A. General rules for the interpretation of the Combined Nomenclature

General rule 5(b) Packing containers normally used for marketing beverages, jam, mustard, spices, etc., are to be classified with the goods they contain even if clearly suitable for repetitive use.

C. General rules applicable both to nomenclature and to duties

General rule 3 In Article 48(2) of Commission Implementing Regulation (EU) 2015/2447 ⁽¹⁾, 'working days' means all days other than Saturdays, Sundays and the published public holidays of the European Central Bank.

⁽¹⁾ Commission Implementing Regulation (EU) 2015/2447 of 24 November 2015 laying down detailed rules for implementing certain provisions of Regulation (EU) No 952/2013 of the European Parliament and of the Council laying down the Union Customs Code (OJ L 343, 29.12.2015, p. 558).

SECTION I

LIVE ANIMALS; ANIMAL PRODUCTS

CHAPTER 1

LIVE ANIMALS

0101	Live horses, asses, mules and hinnies
0101 29 10 and 0101 29 90	Horses Wild horses, such as the Przewalski horse or tarpan (Mongolian horse), are included in these subheadings. However, zebras (<i>Equus zebra</i>, <i>Equus grevyi</i>, <i>Equus burchelli</i>, <i>Equus quagga</i>, etc.) are classified in subheading 0106 19 00, even though they belong to the Equidae family. Hybrids of the mare and zebra (zebrule) are classified in subheading 0106 19 00.
0101 30 00	Asses Both domestic and non-domestic asses fall in this subheading. Among the latter are Mongolian djiggetai, the Tibetan kiang, the onager and the hemione or half-ass (<i>Equus hemionus</i>). Hybrids of the ass and zebra (zebrass) are classified in subheading 0106 19 00.
0101 90 00	Other This subheading covers the animals described in the final paragraph of the HS Explanatory Note to heading 0101.
0102	Live bovine animals
0102 21 10 to 0102 29 99	Cattle These subheadings include the animals described in the HS Explanatory Note to heading 0102, first paragraph, (1). Yaks have 14 pairs of ribs, while all the other animals of the bovine species (with the exception of the European and American bison) have only 13 pairs of ribs.
0102 31 00 to 0102 39 90	Buffalo These subheadings include the animals described in the HS Explanatory Note to heading 0102, first paragraph, (2). The European bison (<i>Bison bonasus</i>) and the American bison (<i>Bison bison</i>) have 14 pairs of ribs, while all the other animals of the bovine species (with the exception of yaks) have only 13 pairs of ribs.
0102 39 10	Domestic species This subheading covers all the animals of the domestic bovine species of the genus <i>Bubalus</i>, irrespective of their intended use (stock, raising, fattening, breeding, slaughter, etc.), but excluding pure-bred breeding animals (subheading 0102 31 00).
0102 39 90	Other This subheading covers the bovine species of the genera <i>Syncerus</i> and <i>Bison</i>, but excluding pure-bred breeding animals (subheading 0102 31 00).
0102 90 20 to 0102 90 99	Other These subheadings include the animals described in the HS Explanatory Note to heading 0102, first paragraph, (3).
0102 90 91	Domestic species This subheading covers all of the animals of the domestic bovine species not included above, irrespective of their intended use (stock, raising, fattening, breeding, slaughter, etc.), but excluding pure-bred breeding animals (subheading 0102 90 20).

CHAPTER 84

**NUCLEAR REACTORS, BOILERS, MACHINERY AND MECHANICAL APPLIANCES;
PARTS THEREOF****General**

For the purposes of subheadings 8407 34 10, 8407 90 50 and 8408 20 10, 'for the industrial assembly' applies only to the series-assembly of new vehicles in factories which manufacture or assemble motor vehicles (including subcontractors).

These subheadings can be applied only to engines actually used in the production of new vehicles of the kinds referred to in the subheadings. Consequently, they do not cover similar engines used as spare parts.

8402 Steam or other vapour generating boilers (other than central heating hot water boilers capable also of producing low pressure steam); superheated water boilers**8402 19 90 Other**

This subheading includes hybrid firetube water-tube boilers and also special tank-boiler variants, such as electrically heated steam boilers equipped with electric-heating cartridges instead of firing plant.

8405 Producer gas or water gas generators, with or without their purifiers; acetylene gas generators and similar water process gas generators, with or without their purifiers**8405 10 00 Producer gas or water gas generators, with or without their purifiers; acetylene gas generators and similar water process gas generators, with or without their purifiers**

This subheading does not include:

- (a) town gas generators (coking ovens), as used in gasworks (heading 8417);
- (b) electrolytic gas generators (for example, for the generation of nitrogen dioxide, hydrogen sulphide or prussic acid, depending upon the electrolyte used) are to be classified in heading 8543.

8407 Spark-ignition reciprocating or rotary internal combustion piston engines**8407 21 10 to 8407 29 00 Marine propulsion engines**

These subheadings do not cover engines used on board water craft for purposes other than propulsion.

8408 Compression-ignition internal combustion piston engines (diesel or semi-diesel engines)**8408 10 11 to 8408 10 99 Marine propulsion engines**

See the explanatory note to subheadings 8407 21 10 to 8407 29 00.

8409 Parts suitable for use solely or principally with the engines of heading 8407 or 8408

In addition to the exclusions referred to in the HS Explanatory Note to heading 8409, the following are also excluded from this heading:

- (a) piping and tubing, of unhardened vulcanised rubber (heading 4009);
- (b) flexible tubing and piping, of base metal (heading 8307);
- (c) gaskets and similar joints (generally, classified according to their constituent material or in heading 8484).

8409 99 00	Other This subheading does not include exhaust-gas turbochargers used to compress the atmospheric air needed for combustion in order to boost the power of internal-combustion piston engines. As turbine-driven air compressors driven by an exhaust-gas turbine, they fall in heading 8414.
8411	Turbojets, turbopropellers and other gas turbines
8411 11 00 to 8411 12 80	Turbojets These subheadings do not cover so-called after-burning auxiliary appliances, presented separately (subheading 8411 91 00).
8411 99 00	Other This subheading includes rotor (fan) blades for gas turbines, whether or not the gas turbine functions as a prime mover in combination with an electric generator.
8412	Other engines and motors
8412 21 20 to 8412 29 89	Hydraulic power engines and motors These subheadings include motors for hydraulic transmission.
8412 21 20 and 8412 21 80	Linear acting (cylinders) These subheadings cover hydraulic positioning and locking actuators used to adjust the position of seats for the crew of aircraft.
8413	Pumps for liquids, whether or not fitted with a measuring device; liquid elevators
8413 50 20	Hydraulic units See the HS Explanatory Note to heading 8412, (B), (6).
8413 60 20	Hydraulic units See the HS Explanatory Note to heading 8412, (B), (6).
8413 70 51 to 8413 70 75	Radial flow pumps In radial flow pumps, the liquid is propelled at right angles to the axis of the impeller.
8413 70 81 and 8413 70 89	Other centrifugal pumps These subheadings include, in particular, axial-flow centrifugal pumps in which the liquid is propelled along the axis of the impeller. They also include hybrid types of radial-flow and axial-flow pumps (for example, radial intake and axial outflow or axial intake and radial outflow).
8413 81 00 and 8413 82 00	Other pumps; liquid elevators These subheadings do not cover: (a) so-called medical suction pumps for drawing off secretions, which, in addition to the pump, also consist of a suction device and are used in operating theatres or emergency accident ambulances (mobile operating theatres) (heading 9018); (b) medical pumps which are to be worn, carried or implanted in the body, fitted with a supply reservoir and acting as medicine dispensers incorporating an energy source to drive the pump, in a common housing (heading 9021).

8414	Air or vacuum pumps, air or other gas compressors and fans; ventilating or recycling hoods incorporating a fan, whether or not fitted with filters For the purposes of this heading, pumps and compressors also include motor pumps, turbopumps, motor compressors and turbo-compressors.
8414 10 89	Other This subheading includes liquid-ring vacuum pumps and membrane vacuum pumps.
8414 20 20 and 8414 20 80	Hand- or foot-operated air pumps Only those pumps which are hand- or foot-operated, i.e., operated solely by human effort, as described in the HS Explanatory Note to heading 8414, (A), fall in these subheadings. They must be designed, in particular, for the inflation of pneumatic tyres (for example, for cycles, motor vehicles, etc.) and similar goods, for example, air mattresses, air cushions and inflatable boats.
8414 51 00 to 8414 59 95	Fans Only those goods displaying the characteristics described in the HS Explanatory Note to heading 8414, (B), and complying with the following conditions are considered to be fans for the purposes of these subheadings: 1. the air or gas pressure must not exceed 2 bar; 2. they must incorporate only one rotating surface (one stage). Air compressors not satisfying the above conditions are classified in subheadings 8414 80 11 to 8414 80 80.
8414 59 35	Centrifugal fans Centrifugal fans are those in which air or other gas to be moved is taken in axially and discharged radially.
8414 80 11 and 8414 80 19	Turbo-compressors In a turbo-compressor, the vane wheel shaft is driven by an external motor and the air or the other gas to be pumped is set in motion by the vane wheel. Turbo-compressors may be single-stage or multi-stage, axial or radial. Simple two-stage compressors are used, for example, in vacuum cleaners.
8414 80 11	Single-stage This subheading also includes exhaust-gas turbochargers for internal-combustion piston engines, where they are used to compress the atmospheric air needed for combustion in order to increase power output. They are single-stage air turbo-compressors (blowers) used to generate an overpressure of more than 2 bar and are driven by an exhaust-gas turbine installed adjacent to the compressor casing. The exhaust-gas turbine is supplied with exhaust gases from the internal-combustion piston engine on which it is mounted.
8414 90 00	Parts This subheading also includes parts of exhaust-gas turbochargers for internal combustion engines. However, parts of the exhaust-gas turbine used with exhaust-gas turbochargers are classified in heading 8411 as parts of a gas turbine without combustion chamber.
8418	Refrigerators, freezers and other refrigerating or freezing equipment, electric or other; heat pumps other than air-conditioning machines of heading 8415
8418 69 00	Other This subheading includes 'chill dryers' which are devices used to dry the air in swimming-pool buildings and other damp premises. Basically, they consist of a refrigerating unit and a motor-driven fan. The fan ingests the moist air, which is then ducted to the refrigerator-unit evaporator, where it is condensed on the cold walls. The resultant condensate water is caught in a trough. The dried air is led for reheating across the heated refrigerator-unit condenser and fed back into the room.

In the case of the chill dryers used to dry compressed air in the compressed-air systems also covered by this subheading, the dried compressed air is normally reheated by means of an additionally installed air-air heat exchanger. This heat exchanger conducts the heat from the damp compressed air entering on the input side of the chill dryer via walls to the dried, compressed air.

These appliances are not provided with devices for adjusting the temperature of the air.

However, this subheading does not cover devices producing dry ices (slab ice) by means of the rapid reduction of the pressure of high-pressure carbon dioxide with resultant supercooling (heading 8479).

8419

Machinery, plant or laboratory equipment, whether or not electrically heated (excluding furnaces, ovens and other equipment of heading 8514), for the treatment of materials by a process involving a change of temperature such as heating, cooking, roasting, distilling, rectifying, sterilising, pasteurising, steaming, drying, evaporating, vaporising, condensing or cooling, other than machinery or plant of a kind used for domestic purposes; instantaneous or storage water heaters, non-electric

8419 20 00**Medical, surgical or laboratory sterilisers**

The apparatus covered by this subheading, intended as equipment for clinics, operating theatres, medical centres, etc., consists of receptacles in which medical and surgical instruments as well as cotton wool, absorbent lint and other dressings are subjected to a temperature of 100 °C or more in order to destroy the micro-organisms they may be carrying.

They are usually in the form of a paralleliped or a cylinder standing on a pedestal and containing removable grids. Generally, the container is of steel or aluminium and lined with insulating material. The door may be glazed to allow the instruments placed in the interior to be seen. Some appliances are in the form of cabinets or other furniture. In this case, a steriliser proper may incorporate a rack for storing instruments or other articles to be sterilised; this will not affect its classification in this subheading.

These appliances are heated by spirit, petroleum, gas or electricity and sterilisation is effected by boiling water (vats), steam under pressure (autoclaves) or hot dry air (ovens), depending on the design of the appliance.

**8419 50 20
and
8419 50 80****Heat-exchange units**

Heat-exchange units are used:

1. to bring about a change of temperature in fluids or gases without changing their state, this change of temperature being in some cases sufficient to cause sterilisation or pasteurisation;
2. to vaporise or condense fluids.

This subheading includes:

1. condensers for nitrogen and other gases;
2. devices called 'freezers', for the cooling and condensing of solvents used mainly in dyeing and dry-cleaning;
3. cooling apparatus for liquids, vapour or gas, used in various industries (for example, dairies, breweries);
4. continuous pasteurisation apparatus, used notably in dairies (plate pasteurisers).

This subheading does not cover:

(a) instantaneous or storage water heaters of subheading 8419 11 00 or 8419 19 00;

(b) apparatus in which:

- the thermal exchange is used to transform a liquid or gaseous substance into a solid state (for example, spray drying);
- the thermal exchange between the two fluids does not take place through a wall (for example, a free air-flow tower).

Such apparatus is generally classified in subheading 8419 89 10 or 8419 89 98.

8419 89 10**Cooling towers and similar plant for direct cooling (without a separating wall) by means of recirculated water**

This subheading covers cooling towers in which the water to be cooled releases its heat directly into the air by means of evaporation. The heated water is pumped up, then trickles down freely inside the cooling tower and in so doing is cooled by the rising air (chimney effect).

8419 89 98**Other**

This subheading covers apparatus for smoking raw sausages, even when the sausage undergoes thermal treatment during smoking which results in the partial or total cooking of the sausage. The apparatus consists of a large chamber heated by steam coils; hot or cold smoke is blown into the chamber from outside by a fan. It is also equipped with a humidifier and with coils containing cold water for cooling. The raw sausages are conveyed into the chamber, suspended on moving frames.

Items of equipment known as 'tableware dispensers' for the holding and dispensing of tableware to diners in canteens or self-service restaurants are excluded from this subheading, even where they are equipped with an electrical heating device or water-bath heating for warming the tableware (heading 9403).

8421**Centrifuges, including centrifugal dryers; filtering or purifying machinery and apparatus, for liquids or gases****8421 39 35****By a catalytic process**

This subheading includes catalytic exhaust gas purifiers for fitting in the exhaust systems of motor vehicles or in flue gas lines of industrial plant, to remove the nitrogen oxide and, possibly, other pollutants (for example, carbon monoxide and hydrocarbons) from the exhaust or flue gases by means of a chemical reaction in order to reduce air pollution. Catalytic converters fitted in motor vehicles consist of a housing containing a ceramic honeycomb (monolithic support) composed of channels coated with activated catalytic material. In industrial plant, flue gas purifiers consist of a support frame containing a large number of catalytic components. Monolithic supports and catalytic components presented separately, however, are classified as catalysts in heading 3815.

8421 39 85**Other**

In addition to apparatus operating by an electrostatic and thermic process, this subheading includes gas-purifying apparatus which separates the gas mixture into its individual components by a retaining process.

8422**Dishwashing machines; machinery for cleaning or drying bottles or other containers; machinery for filling, closing, sealing or labelling bottles, cans, boxes, bags or other containers; machinery for capsuling bottles, jars, tubes and similar containers; other packing or wrapping machinery (including heat-shrink wrapping machinery); machinery for aerating beverages****8422 90 10****Of dishwashing machines**

Programmed time switches for dishwashing machines presented separately are classified according to their characteristics (for example, heading 9107 00 00).

8423**Weighing machinery (excluding balances of a sensitivity of 5 cg or better), including weight-operated counting or checking machines; weighing-machine weights of all kinds****8423 20 10****Using electronic means for gauging weight**

These scales are considered to be 'electronic scales' if microprocessors are built into the scales' measuring instrument, for example, in order to calculate the cost of the amount weighed using the unit price (for example, price per kilogram) entered into the scale.

8423 20 90**Other**

This subheading covers electromechanical scales for conveyors. The design and method of operation of such scales correspond to those of electromechanical scales described in the explanatory note to subheading 8423 81 80.

8423 30 10**Using electronic means for gauging weight**

The explanatory note to subheading 8423 20 10 applies *mutatis mutandis*.

8423 30 90**Other**

This subheading covers electromechanical constant weight scales and scales for discharging a predetermined weight of material into a bag or container, including hopper scales. The design and method of operation of these scales correspond to those of the electromechanical scales described in the explanatory note to subheading 8423 81 80.

**8423 81 21
to
8423 81 29****Using electronic means for gauging weight**

The explanatory note to subheading 8423 20 10 applies *mutatis mutandis*.

8423 81 80**Other**

This subheading covers electromechanical scales in which the weight of objects is converted into an electrical signal (current) by means of a transducer fitted into the scales, the current then being measured by a measuring device incorporated in the scales. As a rule the measuring transducers in such scales consist of load cells or bars with strain gauges (electrical resistances) connected to form an electrical bridge. The force exerted by the mass to be weighed deforms the load cells, causing a change in the length of the strain gauge (contraction or elongation) and thus a change of resistance which is proportional to the mass to be weighed and which is passed in the form of a change of current to the measuring device on the scale by the resistance-measuring bridge.

In addition to the measuring instrument, known as a weighing-out unit, scale display or weighing indicator, which is normally contained in a single casing, electromechanical scales can also have other units which are wired up together (for example, keyboard, memory unit, printer, VDU, controller and card reader for controlling access to the scales). In weighing systems of this kind, several scales may be connected to a common measuring instrument (also known as 'remote weighing terminals').

The above scales may also be equipped with an interface through which the scales can be connected to an automatic data-processing unit.

8423 82 20**Using electronic means for gauging weight, excluding machines for weighing motor vehicles**

The explanatory note to subheading 8423 20 10 applies *mutatis mutandis*.

**8423 82 81
and
8423 82 89****Other**

The explanatory note to subheading 8423 81 80 applies *mutatis mutandis*.

8423 89 20**Using electronic means for gauging weight**

The explanatory note to subheading 8423 20 10 applies *mutatis mutandis*.

8423 89 80**Other**

The explanatory note to subheading 8423 81 80 applies *mutatis mutandis*.

8424**Mechanical appliances (whether or not hand-operated) for projecting, dispersing or spraying liquids or powders; fire extinguishers, whether or not charged; spray guns and similar appliances; steam or sandblasting machines and similar jet projecting machines****8424 30 10****Compressed air operated**

This subheading includes sand-blasting machines incorporating an air compressor used for cleaning sparking plugs or trimming monolithic electrical condensers.

Trimming denotes the removal of the conductive condenser material by applying a jet of sand to attain the desired capacitance in the condenser.

8424 89 70**Other**

This subheading includes 'washing machines' for cleaning motor vehicles, metal parts or other articles by the spraying of water, petrol or other fluids; they are fitted with a pump, hoses with nozzles as well as, where appropriate, a feed device, heating apparatus etc., to form a complete unit.

However, this subheading does not include high-pressure water cleaning appliances (subheadings 8424 30 01 and 8424 30 08).

8426 Ships' derricks; cranes, including cable cranes; mobile lifting frames, straddle carriers and works trucks fitted with a crane**8426 41 00 Other machinery, self-propelled****and
8426 49 00**

See the HS Explanatory Note to heading 8426, 'self-propelled and other "mobile" machines', (b), (2).

In order to distinguish between the self-propelled machinery of these subheadings, on the one hand, and powered vehicles for special purposes of heading 8705, on the other hand, the self-propelled machines falling in these subheadings generally have the following characteristics:

1. they are propelled by an engine forming part of the lifting device;
2. they have a maximum speed of 20 km per hour;
3. they have a single cabin forming part of the lifting device;
4. they do not usually move when laden or, if they do, only when the movement is limited and incidental to their lifting function.

8428 Other lifting, handling, loading or unloading machinery (for example, lifts, escalators, conveyors, teleferics)

This heading does not include caterpillar lifts with caterpillar track systems used to move wheelchairs with their occupants up or downstairs, these fall in heading 8714 as accessories for wheelchairs.

8428 90 90 Other

This subheading includes:

1. roller tables or roller conveyors, power-driven or not, intended for feeding the stock into the rolling stand and removing it therefrom or for transporting the stock from one stand to another;
2. pinch rolls with an analogous function to the above machines, consisting of two series of rollers between which the stock passes;
3. roller cooling beds which are situated at the end of the rolling-mill lines on which the stock (for example, bars and wire) is conveyed slowly while being allowed to cool in the ambient air;
4. lifting or tilting tables for rolling-mills with several superimposed sets of rolls (three high and double two high stands). Such apparatus comprises a table tilting on an axis situated at the furthest extremity of the rolling-mill. The tables are fitted with power-driven rollers. After passing through one set of rollers, the stock arrives on the table which then tilts and aligns with the next set of rolls between which the stock is pushed by means of power-driven rollers;
5. transfers used in parallel-stand rolling-mills for transferring, for example, bars from the roller table of the first stand to that of the second stand;
6. tilting and turn-over devices;
7. machinery designed for the remote handling of highly radioactive substances.

This subheading does not include automatic ingot manipulators and machinery (for example, cranes and overhead travelling cranes), which, although used for feeding rolling-mills, does not participate directly in the work of the latter (subheading 8426 12 00, 8426 41 00, 8426 49 00 or 8426 99 00).

8429 Self-propelled bulldozers, angledozers, graders, levellers, scrapers, mechanical shovels, excavators, shovel loaders, tamping machines and roadrollers**8429 30 00 Scrapers**

See the HS Explanatory Note to heading 8429, second paragraph, (C).

This subheading does not cover combination scrapers, which consist of a tractor (even with only a single axle) and a scraper proper. Each element must, by application of note 2 to Chapter 87, be classified in its own appropriate heading (heading 8701 for the tractor and subheading 8430 69 00 for the scraper).

8432	Agricultural, horticultural or forestry machinery for soil preparation or cultivation; lawn or sports-ground rollers
8432 39 11	Central driven precision spacing seeders This subheading covers machines which place the seed or individual grains in the soil in rows with controlled uniform spacing. These machines can also place seeds simultaneously in several rows.
8433	Harvesting or threshing machinery, including straw or fodder balers; grass or hay mowers; machines for cleaning, sorting or grading eggs, fruit or other agricultural produce, other than machinery of heading 8437
8433 11 10 to 8433 19 90	Mowers for lawns, parks or sports grounds See the HS Explanatory Note to heading 8433, part (A), penultimate paragraph, with reference to riding lawnmowers.
8438	Machinery, not specified or included elsewhere in this chapter, for the industrial preparation or manufacture of food or drink, other than machinery for the extraction or preparation of animal or fixed vegetable fats or oils
8438 80 10	For the preparation of tea or coffee This subheading includes machines for blending various teas and coffee-grinding machines. This subheading does not include machines for the industrial preparation of ready-made tea or coffee (subheading 8419 81 20), coffee-roasting machines (subheading 8419 89 98), machines for manufacturing coffee powder (instant coffee) (subheading 8419 39 00).
8439	Machinery for making pulp of fibrous cellulosic material or for making or finishing paper or paperboard
8439 30 00	Machinery for finishing paper or paperboard In addition to the machinery referred to in the HS Explanatory Note to heading 8439, part III, this subheading also covers machines giving a parchment finish. This subheading does not cover: (a) machines which having analogous functions to those referred to above, operate not on the paper or paperboard sheets, but on products of these materials. For example, machines for waxing paper cups and containers, etc., by immersion (subheading 8479 89 97). Machines which make paper or paperboard products such as paper cups, boxes, containers, etc., fall in heading 8441; (b) machines which, while operating on paper or paperboard in sheets, make products which are no longer paper or paperboard for the purposes of the Combined Nomenclature. This is the case, for example, with machines for coating paper with abrasives or for applying photosensitive emulsions (subheading 8479 89 97).
8441	Other machinery for making up paper pulp, paper or paperboard, including cutting machines of all kinds
8441 10 20	Other slitting and cross-cutting machines This subheading covers machines which produce sheets of a specific format from continuous webs of paper either by cross-cutting only or by slitting and cross-cutting simultaneously.
8441 10 30	Guillotines This subheading covers machines which cut piles of paper with a single blade, while a beam placed above the cutting table holds the paper firmly in place along the cut. This subheading also covers machines with a single knife and a revolving cutting table for trimming book blocks, whereby the book blocks are turned 90° after the first and second cuts.

8441 10 70**Other**

This subheading includes machines with three knives set at 90° to each other which are used for trimming book blocks. Two knives simultaneously trim the top and bottom edges of the book blocks and then the third knife trims the fore-edge or vice versa.

This subheading also includes hand- or pedal-operated machines.

8442**Machinery, apparatus and equipment (other than the machine tools of headings 8456 to 8465) for preparing or making plates, cylinders or other printing components; plates, cylinders and other printing components; plates, cylinders and lithographic stones, prepared for printing purposes (for example, planed, grained or polished)****8442 30 00****Machinery, apparatus and equipment**

This subheading includes phototypesetting and composing machines, which compose by successively photographing characters mounted on revolving discs or the face of special matrices or the characters created on a cathode-ray tube by a matrix of very small overlapping dots. This subheading also includes composing machines using a laser beam projected onto photographic film.

This subheading also includes machines for typefounding and typesetting (for example, linotypes, monotypes, intertypes), with or without founding device.

Type-founding may be a hand operation, or may be done mechanically by means of more or less complex machines. The apparatus and machines covered include:

1. matrices, small plates, usually of copper or nickel, impressed by means of a punch. They are used for casting separate printing types;
2. tables for levelling, by hand planing, the face of printing type. These consist essentially of a perfectly smooth table slit down the centre and fitted with a clamping device which holds the type in place;
3. automatic typefounders. In these, the printing type is made letter by letter, but is not set. They usually consist of an electrically heated crucible containing the molten metal, of a mould cooling device to accelerate the hardening, and of mechanisms for planing and levelling the type;
4. typecasting machines for rules, spacing material (furniture), etc., operating by extrusion;
5. composing sticks (or setting sticks) used to start the composition, and on which one or more lines of type are set by hand. They consist mainly of a small well planed wooden or metal plate, with flanges on two adjacent sides and often with a mobile clamping slide. The subheading includes galleys, similar, but larger, which hold the type for a whole page;
6. chases, cast iron or steel frames to hold several pages for printing. One, two or four pages are held in place in the chase by means of metal quoins (special metal wedges) or by mechanical wedging devices (nut or screw type, etc.) which also fall in this heading.

In addition to the above typefounding and typesetting machines and apparatus, there is a whole group of other machines which mechanically cast the type and also set it. This is done either as two distinct operations on two different but complementary machines (the first machine produces a perforated paper band which controls the second machine whose function is to cast the type either separately or in line blocks), or as one operation on the same machine. These machines, often very complex, include:

1. casting and setting machines for separate types (monotype) which, operating from a band previously perforated on a precomposing machine, select, by means of pneumatic relays, special matrices contained within the machine which produces the individual type characters and sets them in a galley (itself incorporated in the machine).

These machines are used in conjunction with a precomposing machine having a keyboard perforator which produces the precomposition on a paper tape. These precomposing machines are also classified here;

2. keyboard machines for casting and setting separate types, all operations being carried out on the same machine (Rototype, etc.);
3. typefounders for lineset type. The matrices, after being set in lines by hand, are incorporated in the machine which casts the type and delivers it in the form of a line of type (slug);

4. machines for setting and founding lines of type. These are complex keyboard machines of various types (Intertype, Linograph, etc.) which both set and cast the type in the form of lines on the same machine. Some of these machines are fitted with a device to enable them to operate from paper bands previously perforated on a separate machine which is also covered by this heading.

This subheading covers the machinery, apparatus and equipment for preparing or making plates, cylinders or other printing components described in the HS Explanatory Note to heading 8442, (A), (1) to (3).

This subheading does not, however, cover:

- (a) general-purpose automatic data-processing machines which are used for photocomposition in addition to having bookkeeping and payroll accounting functions, for example, and for this purpose are connected by cable (on-line) to a phototypesetting and composing machine in order to control it and supply it with type-setting data (heading 8471 — see note 5(A) to this chapter);
- (b) stand-alone telegraphic transmitter and receiver apparatus (heading 8517).

8442 50 00**Plates, cylinders and other printing components; plates, cylinders and lithographic stones, prepared for printing purposes (for example, planed, grained or polished)**

Natural lithographic stone consists of a kind of limestone with fine homogeneous grains. Artificial lithographic stone is usually made from cement and calcium carbonate which has been ground and compressed.

Lithographic stones of these subheadings are:

- covered with illustrations, either hand drawn or photographically transferred; or
- planed or grained so as to be rendered suitable, without further treatment, for drawings or writing.

Limestone (known as 'lithographic stone') in the crude state is excluded from these subheadings and falls in subheading 2530 90 00.

8443**Printing machinery used for printing by means of plates, cylinders and other printing components of heading 8442; other printers, copying machines and facsimile machines, whether or not combined; parts and accessories thereof****8443 39 00****Other**

This subheading includes blueprinters and diazocopiers which are used for copying translucent originals onto photosensitive paper. As a result of light passing through the translucent original, the diazo compound or the photosensitive iron salts in the copying paper are broken down in the illuminated areas. The non-illuminated areas are rendered visible by developing. This apparatus normally provides bluish copies, in which the individual lines are not quite as sharp as the original.

This subheading also includes contact type machines and thermocopying apparatus.

This subheading includes non-impact printers, provided that they are not capable of connecting to an automatic data-processing machine or to a network, such as:

1. thermal printers, in which an electrically heated thermal print head produces the desired characters on heat-sensitive paper by means of a dot matrix;
2. electrostatic printers, in which the moving, statically charged metal tips of the write head produce invisible characters made up of small electrostatically charged dots on electrographic paper. Toner fluid is used to ink the charged dots and produce visible characters.

The printers listed above are controlled by data media (for example, CD ROM, diskettes, magnetic tapes or semiconductor media) or by machines other than automatic data-processing machines (for example, digital cameras, video camera recorders or telephones for cellular networks).

8443 91 10
to
8443 99 90

Parts and accessories

In addition to the parts and accessories mentioned in the HS Explanatory Note to heading 8443, these subheadings include:

1. automatic feeders for feeding tinplate printing machines;
2. tensioning systems, usually pneumatic, for maintaining the tension of the paper drawn from the reel holders;
3. anti-offsetting devices, other than those of the spray type;
4. print chains, pin-heads, type heads and daisy wheels.

8443 99 10

Electronic assemblies

Electronic assemblies consist of one or more printed circuits fitted with electronic integrated circuits of heading 8542. They may also contain discrete active elements, discrete passive elements, articles of heading 8536 or other electrical or electromechanical devices, provided that they do not lose their character as electronic assemblies.

This subheading does not include:

- (a) mechanisms (without electronic components);
- (b) modules (composed of mechanisms and electronic assemblies), such as cassette, compact disc or DVD drives for data, video or audio reproduction consisting of mechanisms and electronic control and signal processing assemblies.

8445

Machines for preparing textile fibres; spinning, doubling or twisting machines and other machinery for producing textile yarns; textile reeling or winding (including weft-winding) machines and machines for preparing textile yarns for use on the machines of heading 8446 or 8447

8445 90 00

Other

In addition to the machines described in the HS Explanatory Note to heading 8445, (E), this subheading also includes machines for assembling warp yarn on the beam from warper drums, machines for interlacing and supplying the thread during weaving and threading machines for embroidery.

'Hand knotters', on the other hand, which are small tools held in the hand for knotting broken threads, fall in heading 8205.

8446

Weaving machines (looms)

This heading covers the weaving machines described in the HS Explanatory Note to heading 8446, including Axminster carpet looms, chenille looms, pile fabric looms, terry fabric looms, crepe looms, linen looms, sail looms, transmission belt looms and strap looms.

8447

Knitting machines, stitch-bonding machines and machines for making gimped yarn, tulle, lace, embroidery, trimmings, braid or net and machines for tufting

8447 20 20
and
8447 20 80

Flat knitting machines; stitch-bonding machines

These subheadings include crochet machines which are, in fact, chain knitting machines used for making ornamental trimmings, fringes, curtains, net, lace, etc. (such as galloon braiding looms, lace frames, curtain machines and ribbon looms).

8447 90 00

Other

In addition to the machines described in the HS Explanatory Note to heading 8447, (C), this subheading includes:

1. hand embroidery machines (embroidery machines with pantograph shuttles), automatic shuttle embroidery machines, equipped with jacquard or similar mechanisms, embroidery machines with a number of needles, and multi-head automatic embroidery machines (with several machine heads on one table and equipped with a jacquard or similar mechanism);
2. bobbin lace machines, which produce lace consisting of one or several threads (bobbin lace) using bobbins;

3. braiding and spindle machines, which, by interlacing threads by means of feeders (spindles) equipped with bobbins of thread and making circular and sinusoidal trajectories, produce articles in the piece (for example, flat or round braids) or in shapes (piping, braids, rovings, tubular braids, piping for garments), or cover buttons, wooden articles, fluting, etc., with threads (for example, round braid looms, tubular plait looms, packing braid looms).

However, special braiding machines intended for covering cable wire or other electrical conductors or which braid or twist flexible wire conductors, fall in heading 8479;

4. machines for making trimmings (other than the braid-making machines referred to in 3 above), including:
- (a) machines for making pompons and chenille;
 - (b) machines for making round and fancy chenille and machines for making chenille garlands for Christmas trees;
 - (c) fringe twisting and cutting machines.

However, balancing machines, i.e., weaving machines for making ribbons or trimmings, fall in heading 8446 and crochet machines for making trimmings fall in subheading 8447 20 20 or 8447 20 80.

8448

Auxiliary machinery for use with machines of heading 8444, 8445, 8446 or 8447 (for example, dobbies, jacquards, automatic stop motions, shuttle changing mechanisms); parts and accessories suitable for use solely or principally with the machines of this heading or of heading 8444, 8445, 8446 or 8447 (for example, spindles and spindle flyers, card clothing, combs, extruding nipples, shuttles, healds and heald-frames, hosiery needles)

8448 11 00 and 8448 19 00

Auxiliary machinery for machines of heading 8444, 8445, 8446 or 8447

These subheadings do not include:

- (a) machines for removing the remaining threads from the quills of weaving machines and blade cleaners (subheading 8479 89 97);
- (b) equipment for checking the regularity of threads by rolling onto a drum or board (subheading 9017 80 90);
- (c) thread cleaners using electronic processes — photoelectric cells, for example (subheading 9031 80 20 or 9031 80 80).

8451

Machinery (other than machines of heading 8450) for washing, cleaning, wringing, drying, ironing, pressing (including fusing presses), bleaching, dyeing, dressing, finishing, coating or impregnating textile yarns, fabrics or made-up textile articles and machines for applying the paste to the base fabric or other support used in the manufacture of floor coverings such as linoleum; machines for reeling, unreeling, folding, cutting or pinking textile fabrics

8451 90 00

Parts

This subheading does not include:

- (a) special cones and bobbins for dyeing machinery and apparatus (classified according to their constituent material, see note 1(c) to Section XVI);
- (b) blades and cutters for cropping machines (heading 8208);
- (c) electrostatic devices for flocking machines (heading 8543).

8452

Sewing machines, other than book-sewing machines of heading 8440; furniture, bases and covers specially designed for sewing machines; sewing machine needles

8452 10 11 and 8452 10 19

Sewing machines (lock-stitch only), with heads of a weight not exceeding 16 kg without motor or 17 kg including the motor; sewing machine heads (lock-stitch only), of a weight not exceeding 16 kg without motor or 17 kg including the motor

1. These subheadings cover sewing machines and sewing machine heads having both the following characteristics:
- (a) they must be lock-stitch only (running-stitch, zig-zag stitch, decorative stitch), i.e., operate with two threads, one inserted on the needle and one carried underneath by a shuttle;

- (b) the head must not exceed 16 kg in weight without motor or 17 kg, including the motor (in the case of an incomplete head to be classified as complete in accordance with general interpretative rule 2(a) of the Combined Nomenclature, it is the weight in the complete state which counts).
2. The sewing machine head comprises the working unit of the machine (including, where relevant, the motor incorporated in or mounted on the head) and consists essentially of an arm containing the needle-driving mechanism and a base containing the shuttle mechanism and possibly a device for lifting the presser foot. The frame, table, cover and other furniture (including the pedal board) do not form part of the head.
3. It should be noted, however, that on some portable sewing machines the base is designed to serve also as a stand. In this case, it is regarded as a sewing machine and not as a sewing machine head.

8456 Machine tools for working any material by removal of material, by laser or other light or photon beam, ultrasonic, electrodischarge, electrochemical, electron beam, ionic-beam or plasma arc processes; water-jet cutting machines

8456 11 10 Of a kind used solely or principally for the manufacture of printed circuits, printed circuit assemblies, parts of heading 8517, or parts of automatic data processing machines

This subheading includes machines for trimming electrical resistors in printed circuits using a laser beam. The conductive material which constitutes the resistors on the insulating substrate of the printed circuits is removed by means of a laser beam until the desired resistance is attained.

8456 30 11 Wire-cut

These machines are equipped with an electrode which consists of a small-diameter wire which is wound between two spools set on opposite sides of the workpiece.

8457 Machining centres, unit construction machines (single station) and multi-station transfer machines, for working metal

8457 10 10 Horizontal

This subheading covers machining centres where the tooling is set exclusively in a horizontal spindle and cuts the workpiece from the side.

8457 10 90 Other

This subheading includes machining centres where the tooling is applied to the workpiece from above (vertical spindle machining centres), those using both vertical and horizontal spindles (combined machining centres) or those working through the use of a pivoting head (universal machining centres).

8458 Lathes (including turning centres) for removing metal

This heading covers lathes specially designed for working metals and which operate by removing the material with the help of turning tools. In general, the article to be shaped is driven in a rotary movement around its own axis. However, lathes in which the tool itself or the tool and the article are rotated are also covered by this heading.

In addition to the lathes referred to in the HS Explanatory Note to heading 8458, third paragraph, this heading includes special relieving lathes for axles, bark-stripping lathes and spindle or axle turning machines, as well as universal lathes. These universal lathes resemble slide lathes but are differentiated by a particular mechanism which enables them to perform other functions such as drilling, milling and cutting into lengths, in addition to turning.

8459 Machine tools (including way-type unit head machines) for drilling, boring, milling, threading or tapping by removing metal, other than lathes (including turning centres) of heading 8458

8459 10 00 Way-type unit head machines

See the HS Explanatory Note to heading 8459, third paragraph, (1).

8459 31 00
and
8459 39 00

Other boring-milling machines

See the HS Explanatory Note to heading 8459, third paragraph, (3), third subparagraph.

8460

Machine tools for deburring, sharpening, grinding, honing, lapping, polishing or otherwise finishing metal or cermets by means of grinding stones, abrasives or polishing products, other than gear cutting, gear grinding or gear finishing machines of heading 8461

8460 12 00
and
8460 19 00

Flat-surface grinding machines

These subheadings include the flat-surface grinding machines referred to in the HS Explanatory Note to heading 8460, third paragraph, (3).

These machines are equipped with measuring or setting devices. These devices include:

1. direct reading linear instruments, for example, slide rules and verniers, on which the space between two successive graduations corresponds to a movement of an element of the machine not exceeding 1/100th mm (0,01 mm);
2. profile projectors for checking work in progress. These projectors consist of a graduated ground-glass screen on which a greatly enlarged image of the article and tool is projected so that any travel of the article or tool, as well as the progress of the working process, can be assessed by reference to the graduations on the screen. The work can also be checked by placing on the screen a drawing of the article to be obtained traced on a transparent sheet on a scale corresponding to the magnification factor of the profile projectors; in this case, the machine operator's task is to make the image on the article coincide with the drawing seen on the screen;
3. devices limiting the travel of the tool holder or workpiece holder by means of an adjustable device whose positioning is controlled by means of a slip gauge;
4. electronic checking and control devices for honing and lapping machines which are adjusted by means of a graduated scale on the actual final dimensions of the article and which allow the tool to be slowed down and stopped when the dimensions of the article approach and attain the previously determined dimensions.

8460 22 00
to
8460 29 90

Other grinding machines

See the explanatory note to subheadings 8460 12 00 and 8460 19 00.

8460 22 00

Centreless grinding machines, numerically controlled

Centreless grinding machines are used for grinding the external diameters of workpieces. In the centreless process, the workpiece is not fixed but rests on a rail between the rotating grinding disc and a guide disc (also driven). The diameter required is determined by the distance between the grinding disc and the guide disc.

8460 23 00

Other cylindrical grinding machines, numerically controlled

This subheading includes internal cylindrical grinding machines used for grinding internal diameters of hollow workpieces (for example, the bore). The workpiece is held in a chuck and worked with a grinding wheel small enough to enter the hollow bore of the workpiece.

8460 24 00

Other, numerically controlled

This subheading includes universal grinding machines. These are a combination of external and internal grinding machines and can therefore work simultaneously on both the external and internal diameters of a workpiece.

8460 29 10

For cylindrical surfaces

For internal cylindrical grinding machines, the explanatory note to subheading 8460 23 00 applies, *mutatis mutandis*.

For centreless grinding machines and universal grinding machines, the explanatory notes to subheading 8460 22 00 and to subheading 8460 24 00 apply, *mutatis mutandis*.

8461	Machine tools for planing, shaping, slotting, broaching, gear cutting, gear grinding or gear finishing, sawing, cutting-off and other machine tools working by removing metal or cermets, not elsewhere specified or included
8461 30 10 and 8461 30 90	Broaching machines These subheadings cover broaching machines (see the HS Explanatory Note to heading 8461, third paragraph, (4)); these are machine tools consisting of a multiple-toothed cutting tool, called a broach, used for working the interior or exterior surfaces of an article. In these machines, the article is fixed and the broach, secured to the slide, is pulled or pushed across the work piece in an even, linear cutting motion. The internal broach works and calibrates the internal surfaces of a rough or pre-worked article traversed by the tool. The external broach produces plane or profile surfaces.
8461 40 11 and 8461 40 19	For cutting cylindrical gears Only gears which are obtained from cylindrical blanks and which retain that form after cutting are considered as cylindrical gears for the purposes of these subheadings. These subheadings include machines for making straight-toothed gear wheels, worm gear wheels, worm gears, ratchet wheels and sprocket wheels.
8461 40 31 and 8461 40 39	For cutting other gears These subheadings include machines for making racks, helical gears and worm gears other than cylindrical gears.
8461 40 71 and 8461 40 79	Fitted with a micrometric adjusting system, in which the positioning in any one axis can be set up to an accuracy of at least 0,01 mm See the explanatory note to subheadings 8460 12 00 and 8460 19 00 for measuring and setting devices.
8470	Calculating machines and pocket-size data-recording, reproducing and displaying machines with calculating functions; accounting machines, postage-franking machines, ticket-issuing machines and similar machines, incorporating a calculating device; cash registers
8470 10 00	Electronic calculators capable of operation without an external source of electric power and pocket-size data-recording, reproducing and displaying machines with calculating functions This subheading includes, for example: 1. electronic calculators incorporating a time-and-date display and an alarm or, for example, in addition, a stopwatch device, timer or musical keys; 2. pocket-sized, battery-powered electronic machines which, in addition to performing arithmetical operations, have a memory that can store data for a telephone directory, scheduler, memo pad, calendar, etc. (sometimes known as 'digital diaries'); 3. small, pocket-sized, electronic devices (also called 'minicomputers'), which can be used to form words and sentences which are translated into a chosen foreign language, depending on the memory modules used with these devices; they can also perform simple calculations. They have an alphanumeric keyboard and a rectangular display. This subheading does not cover: (a) wrist watches and pocket watches incorporating a small electronic calculator (heading 9101 or 9102); (b) cigarette lighters incorporating a small electronic calculator and also possibly an electronic timepiece (heading 9613). (c) similar devices without calculating functions (heading 8543).

8470 30 00 Other calculating machines

This subheading includes non-electronic calculating machines mentioned in the HS Explanatory Note to heading 8470, part (A), which for the purposes of calculation employ mechanical devices, generally rack gearing, irrespective of whether they are operated by hand, motor or electromagnetically.

8470 90 00 Other

In addition to the machines listed in the HS Explanatory Note to heading 8470, parts (B) and (D), this subheading includes machines for issuing labels showing the selling price calculated from the weight and unit price, and machines used for issuing motorway toll tickets which at the same time total the amounts involved.

This subheading does not include separately presented printers (heading 8443).

8471 Automatic data-processing machines and units thereof; magnetic or optical readers, machines for transcribing data onto data media in coded form and machines for processing such data, not elsewhere specified or included**8471 70 30 Optical, including magneto-optical**

See additional note 2 to this chapter.

8472 Other office machines (for example, hectograph or stencil duplicating machines, addressing machines, automatic banknote dispensers, coin-sorting machines, coin-counting or -wrapping machines, pencil-sharpening machines, perforating or stapling machines)

This heading includes typewriters fitted with a special device for typing address plates in stencil.

8472 90 90 Other

In addition to the machines described in the HS Explanatory Note to heading 8472, fifth paragraph, (2), (3), (5), (7) to (11), and (16) to (22), this subheading includes:

1. machines for the double-sided coating (facing) of contracts, data sheets, plans, identity cards or other documents with transparent film, in order to protect these documents from aging, falsification, soiling or crumpling;

However, similar machines not used in offices (so-called heat-seal presses), which employ heat and pressure for the purpose of applying thermoplastic transparent film made of synthetic material to the surface or reverse side of pictures, photographs, art prints or other printed products, are excluded from this subheading and fall in heading 8477;

2. electric engraving machines as used, for example, in drawing offices.

This subheading does not include:

- (a) printers of heading 8443;
- (b) pocket-sized, battery-powered electronic machines which, in addition to performing arithmetical operations, have a memory that can store data for a telephone directory, scheduler, memo pad, calendar, etc. (sometimes known as 'digital diaries') (subheading 8470 10 00).

8473 Parts and accessories (other than covers, carrying cases and the like) suitable for use solely or principally with machines of headings 8470 to 8472

In addition to the parts, assemblies and accessories mentioned in the HS Explanatory Note to heading 8473, this heading includes:

1. magnetic heads in Winchester or thin-film technology for peripheral magnetic disk storage units, whether or not incorporated on support arms or in housings;
2. data-storage assemblies (so-called Head/Disk/Assemblies (HDAs)) designed for permanent installation in disk storage units and consisting of several magnetic disks rigidly mounted on a support spindle, of data-head arms with read/write heads, of control, access and positioning mechanisms, incorporated in a single hermetically sealed housing;
3. print and write cassettes, made up in the form of clearly interchangeable accessories and containing, in a special housing, both ribbons and correction tapes.

8473 21 10	Electronic assemblies See the explanatory note to subheading 8443 99 10.
8473 29 10	Electronic assemblies See the explanatory note to subheading 8443 99 10.
8473 30 20 and 8473 30 80	Parts and accessories of the machines of heading 8471 These subheadings do not cover keyboards for automatic data-processing machines, in their own housing (subheading 8471 60 60).
8473 30 20	Electronic assemblies See the explanatory note to subheading 8443 99 10.
8473 30 80	Other parts and accessories of the machines of heading 8471 This subheading does not include cases, whether or not with stands, for tablet Computers (tablet) or mini-tablet Computers (mini-tablet) (heading 4202 or, if they do not have a front cover, according to their constituent material). Such cases are principally designed to protect the back, the sides and the front of a tablet or a mini-tablet and are, therefore, not considered an accessory of the machines of heading 8471 because they do not increase the range of operations of a tablet or a mini-tablet nor do they perform a particular service relative to the main function of the machine.
8473 40 10	Electronic assemblies See the explanatory note to subheading 8443 99 10.
8473 40 80	Other The types of address plates covered by this subheading are those that are used in addressing machines and that contain the addresses to be duplicated, namely by embossing, typing or stamping. Consequently, the plates in question may be made of metal or synthetic material or may take the form of small mounted plates (cards, stencils) made of paperboard, etc. This subheading also covers address plates that have not yet been embossed, typed on or stamped, provided that they are recognizable as address plates for addressing machines. However, goods of this type made of paper or paperboard, i.e., small stencils which are mounted in paperboard frames and which can therefore be inserted in the addressing machines, are covered by heading 4816.
8473 50 20	Electronic assemblies See the explanatory note to subheadings 8443 99 10.
8474	Machinery for sorting, screening, separating, washing, crushing, grinding, mixing or kneading earth, stone, ores or other mineral substances, in solid (including powder or paste) form; machinery for agglomerating, shaping or moulding solid mineral fuels, ceramic paste, unhardened cements, plastering materials or other mineral products in powder or paste form; machines for forming foundry moulds of sand
8474 80 90	Other This subheading includes 3D printers for making articles from the materials referred to in the heading 8474.

8477**Machinery for working rubber or plastics or for the manufacture of products from these materials, not specified or included elsewhere in this chapter**

In addition to the machines mentioned in the HS Explanatory Note to heading 8477, this heading includes:

1. machines for removing rubber soles and rubber heels, which, although used in the shoe industry, are not intended for working leather, skins or hides;
2. machines for cutting blocks of foam plastic, foam rubber, latex foam and similar materials, using a band knife or saw blade.

8479**Machines and mechanical appliances having individual functions, not specified or included elsewhere in this chapter****8479 40 00****Rope- or cable-making machines**

This subheading includes:

1. machines and appliances for making string, cord, rope or cable from textile fibre, such as:
 - (a) stranding machines, which make strands by twisting two or more single threads (cable yarn);
 - (b) rope or cable twisting machines, which twist two or more strands to make large-diameter rope or cable;
 - (c) combined stranding and rope or cable twisting machines, chiefly used for making heavier cord and smaller-diameter rope or cable;
2. machines and appliances for making wire ropes and cables, other than electric cables, working on the same principle as the machines and appliances described in item 1 above;
3. machines and appliances for cabling or braiding electric wires and machines for making coaxial cables.

The subheading does not include machines carrying out operations prior to stranding, for example, combing, spreading, doubling, successive drawing or spinning, or twisting frames of a type used in spinning mills; some machines of this type can also be used for making fine cord by twisting rope yarn (heading 8445).

8479 89 97**Other**

This subheading includes:

1. garage door opening systems, which are mechanical devices for automatically opening and closing up-and-over garage doors by radio remote control. They are fitted to the garage ceiling and consist chiefly of an electric servomotor with drive spindle, a connecting rail with power-transmitting device and an operating arm fixed to the garage door. The servomotor is connected by a cable to the radio RC receiver, which switches on the power supply to the servomotor when it receives control signals from the radio RC transmitter in the vehicle. The radio transmitters and receivers necessary for the remote control of these systems are, however, excluded from this subheading and are classified in heading 8526 in accordance with their own characteristics;
2. certain machines and appliances for the manufacture of printed circuit boards (by machining and cutting larger sheets of paper-base laminate, glass-fibre fabric, ceramic or other insulating material), such as:
 - (a) brushing machines and ultrasonic washing machines for cleaning the sheets of insulating material;
 - (b) rolling or pouring machines for applying photoresist, bonding agent or adhesive to the sheets of insulating material;
3. printed circuit board assembly machines for mounting active, passive or connecting components on printed circuits (pick-and-place machines). These components are automatically fed to the machine on tapes (belts). The machines position the components exactly at the intended locations and mount them on the printed circuit. After the components have been mounted, they are fixed to the printed circuit by, for example, soldering, contact bonding. Apart from the assembly of semiconductor devices, these machines are also able to pick and place other components on substrates.

This subheading does not include machines and apparatus of a kind used solely or principally for the manufacture of semiconductor boules or wafers, semiconductor devices, electronic integrated circuits or flat panel displays (heading 8486).

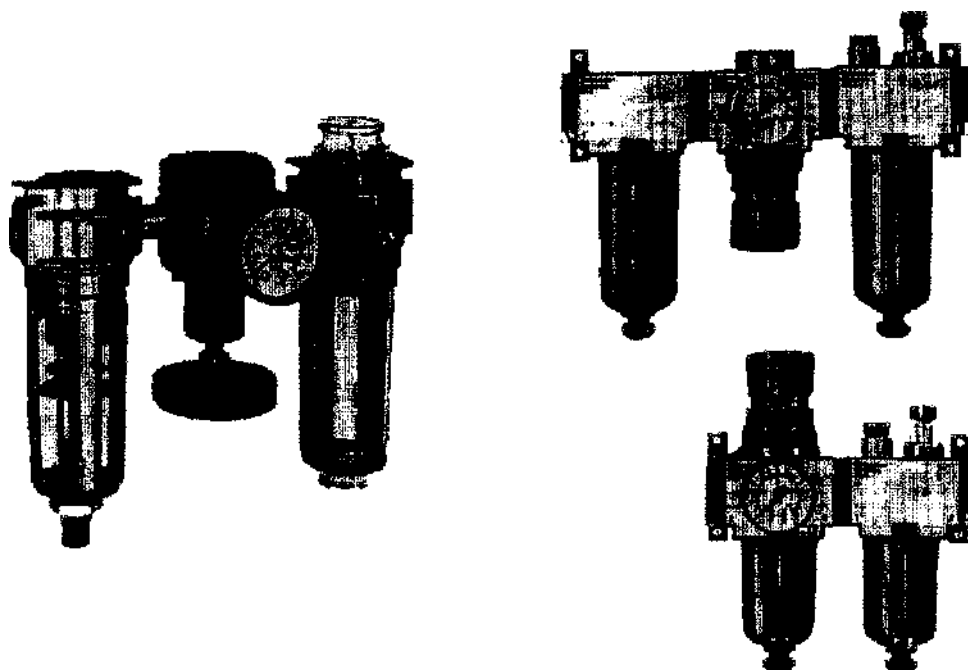
8481 Taps, cocks, valves and similar appliances for pipes, boiler shells, tanks, vats or the like, including pressure-reducing valves and thermostatically controlled valves

8481 10 05

Combined with filters or lubricators

This subheading covers articles consisting of different components which perform the functions needed to regulate compressed-air systems: air filtration (to remove impurities, for example, water, rust, dirt, etc.), regulation to the correct working pressure, lubrication (to ensure smooth operation of the pneumatic components).

They typically have the following appearance:



8486 Machines and apparatus of a kind used solely or principally for the manufacture of semiconductor boules or wafers, semiconductor devices, electronic integrated circuits or flat panel displays; machines and apparatus specified in note 9(C) to this chapter; parts and accessories

This heading excludes, inter alia:

- (a) machines and appliances for the manufacture of printed circuit boards as defined in note 5 to Chapter 85;
- (b) electrical machines and devices for testing wafers or integrated circuits by electrical measurement and for detecting and locating defects, including those equipped with a device for defect marking or a sorting device for assigning the products tested to specific storage containers (Chapter 90).

8486 20 00

Machines and apparatus for the manufacture of semiconductor devices or of electronic integrated circuits

See the HS Explanatory Note to heading 8456, third paragraph.

In addition to the machines mentioned in the HS Explanatory Note to heading 8486, part (B), this subheading includes:

- 1. machine tools for scribing or scoring semiconductor wafers;
- 2. apparatus for rapid heating of semiconductor wafers;
- 3. machine tools for working any material, whereby the removal of material is by means of laser or other light- or photon-beam processes;
- 4. appliances for drying printed or washed insulating substrate and continuous dryers for wafers.

8486 40 00**Machines and apparatus specified in note 9(C) to this chapter**

In addition to the machines mentioned in the HS Explanatory Note to heading 8486, part (D), this subheading includes:

1. die attach apparatus and tape automated bonders for the assembly of semiconductor devices or electronic integrated circuits;
2. photographic cameras, being pattern-generating apparatus of a kind used for producing masks or reticles from photoresist-coated substrates;
3. marking-out instruments, being pattern-generating apparatus of a kind used for producing masks or reticles from photoresist-coated substrates;
4. certain machines and appliances for the manufacture of hybrid integrated circuits (by machining and cutting larger sheets of ceramic or other insulating material), such as:
 - (a) brushing machines and ultrasonic washing machines for cleaning the sheets of insulating material;
 - (b) rolling or pouring machines for applying photoresist, bonding agent or adhesive to the sheets of insulating material.

CHAPTER 85

ELECTRICAL MACHINERY AND EQUIPMENT AND PARTS THEREOF; SOUND RECORDERS AND REPRODUCERS, TELEVISION IMAGE AND SOUND RECORDERS AND REPRODUCERS, AND PARTS AND ACCESSORIES OF SUCH ARTICLES**8501 Electric motors and generators (excluding generating sets)**

This heading includes rotary electric windscreen-wiper motors, without arms or wiper blades but with appropriate transmission mechanisms (spur gear and oscillating connecting rod) which convert the rotary motion into an oscillating one.

8502 Electric generating sets and rotary converters

In addition to the electrical machinery described in the HS Explanatory Note to heading 8502, parts (I) and (II), this heading includes cascade converters, Ward-Leonard sets and rotary dephasers.

8502 39 20 Turbo-generators

Turbo-generators are driven directly by gas or steam turbines. They have a solid cylindrical rotor with longitudinal grooves in which the induction windings are mounted. The rotor may be in one piece or consist of a number of solid sections.

Turbo-generators are generally air-cooled, but high-capacity types are hydrogen-cooled.

8503 00 Parts suitable for use solely or principally with the machines of heading 8501 or 8502

This heading does not include branch circuit plates for electric motors (heading 8536).

8504 Electrical transformers, static converters (for example, rectifiers) and inductors

In addition to the products referred to in the HS Explanatory Note to heading 8504, this heading includes:

1. adjustable transformers (for example, cursor transformers) and variable ratio transformers;
2. field dispersion transformers for fluorescent tubes;
3. special transformers for communications;
4. compensation coils;
5. output coils;
6. smoothing inductors;
7. moving core coils for modifying inductance;
8. ballasts (chokes) for electrical discharge lamps and tubes;
9. Pupin coils;
10. Godefroy coils;
11. stabilised power packs (rectifier with a regulator).

This heading also includes selenium rectifying elements, whether single (selenium plates, in particular) or multiple.

However, this heading does not include silicon or germanium single-crystal rectifiers constituting discrete components (for example, rectifier diodes) or rectifiers constituting integrated circuits, especially microcircuits, whether or not equipped with cooling or insulating devices, etc. Note 8 to this chapter assigns such components to heading 8541 or to heading 8542 (see also note 2 to this chapter).

Furthermore, this heading does not include:

- (a) commutators for multiple-point transformers (heading 8536);
- (b) rectifying lamps, tubes and valves such as phanotrons, thyratrons, ignitrons and high-voltage rectifying tubes for X-ray equipment (subheading 8540 89 00);
- (c) voltage regulators of heading 9032.

8504 40 30**Of a kind used with telecommunication apparatus, automatic data-processing machines and units thereof**

This subheading includes static converters for telecommunication apparatus or for automatic data-processing machines and units thereof, which:

- generally, have stabilising circuits;
- have a typical output voltage of, for example, 3,3, 5, 12, 24, 48 or 60 V.

Static converters for telecommunication apparatus or for automatic data-processing machines and units thereof serve to convert, for example, the alternating current (AC) drawn from the mains supply into the requisite direct current (DC).

Used with automatic data-processing machines, a so-called Uninterruptible Power Supply (UPS) ensures a 'back-up' of energy (with a 'power good' signal) in the event of a power failure, thus preventing data loss.

8504 90 05**Electronic assemblies of machines of subheading 8504 50 20**

For the definition of the expression 'electronic assemblies', see the explanatory note to subheading 8443 99 10.

8504 90 91**Electronic assemblies of machines of subheading 8504 40 30**

For the definition of the expression 'electronic assemblies', see the explanatory note to subheading 8443 99 10.

8505

Electromagnets; permanent magnets and articles intended to become permanent magnets after magnetisation; electromagnetic or permanent magnet chucks, clamps and similar holding devices; electromagnetic couplings, clutches and brakes; electromagnetic lifting heads

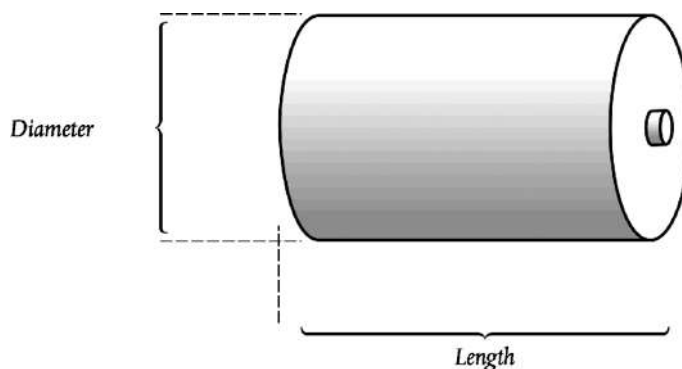
8505 90 29**Other**

This subheading includes electric operating magnets fitted to the doors of private cars and forming part of a central door-locking system. Such systems are connected to the vehicle power supply and are actuated by control signals from an electrical control device belonging to the system. If one of the doors is opened or closed manually, the other vehicle doors will be simultaneously locked or unlocked electromagnetically.

However, this subheading does not include solenoid-operated injection valves for spark-ignition or compression-ignition internal combustion piston engines, the valve bodies and valve needles of which are equipped with a magnet winding and magnet keeper respectively (subheading 8409 91 00 or 8409 99 00).

8506**Primary cells and primary batteries****8506 10 11****Cylindrical cells**

Cylindrical cells have a circular cross-section. Positive and negative poles are at opposite ends. The length of cylindrical cells is greater than their diameter:

**8506 10 91****Cylindrical cells**

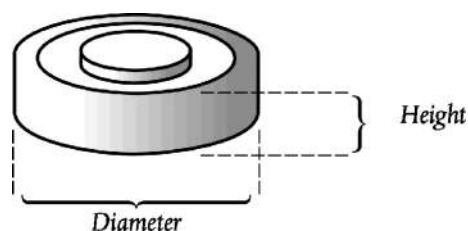
See the explanatory note to subheading 8506 10 11.

8506 50 10**Cylindrical cells**

See the explanatory note to subheading 8506 10 11.

8506 50 30**Button cells**

The height of button cells is less than or equal to the diameter:

**8507****Electric accumulators, including separators therefor, whether or not rectangular (including square)****8507 20 20
and
8507 20 80****Other lead-acid accumulators**

With the exception of accumulators of a kind used for starting piston engines (see subheadings 8507 10 20 and 8507 10 80), these subheadings cover the electric lead-acid accumulators mentioned in the HS Explanatory Note to heading 8507, third paragraph, (1).

These accumulators are used, for example, for driving electric vehicles and for supplying power to telecommunication installations.

**8507 30 20
and
8507 30 80****Nickel-cadmium**

The accumulators of these subheadings are used particularly in safety lamps for miners and often as replacements for dry batteries in equipment such as portable radios, television sets, electric razors and other electrical equipment.

8507 80 00**Other accumulators**

This subheading includes silver-zinc and silver-cadmium accumulators.

**8507 90 30
and
8507 90 80****Parts**

These subheadings do not include connecting parts of accumulator elements (subheading 8536 90 95).

8509**Electromechanical domestic appliances, with self-contained electric motor, other than vacuum cleaners of heading 8508****8509 80 00****Other appliances**

This subheading includes electrical manicure and pedicure sets with self-contained electric motor, connected via a cable to the power-supply unit (mains unit) belonging to the appliance.

8510**Shavers, hair clippers and hair-removing appliances, with self-contained electric motor****8510 10 00****Shavers**

Shavers incorporating hair clippers fall in this subheading.

8511	Electrical ignition or starting equipment of a kind used for spark-ignition or compression-ignition internal combustion engines (for example, ignition magnetos, magneto-dynamos, ignition coils, sparking plugs and glow plugs, starter motors); generators (for example, dynamos, alternators) and cut-outs of a kind used in conjunction with such engines
8511 40 00	Starter motors and dual purpose starter-generators The equipment of this subheading usually operates on 6, 12 or 24 V and has special devices for attachment to motors. This subheading includes: 1. starter motors with sliding armature, starter motors with sliding feeder, starter motors with a helicoidal motion, and starter motors with a sliding pinion and a helicoidal motion; 2. equipment which is the result of combining in one unit a starter motor and a generator.
8512	Electrical lighting or signalling equipment (excluding articles of heading 8539), windscreen wipers, defrosters and demisters, of a kind used for cycles or motor vehicles
8512 30 10	Burglar alarms of a kind used for motor vehicles See the HS Explanatory Note to heading 8512, second paragraph, (11).
8512 90 90	Other This subheading includes frames and reflectors for headlights, and wiper arms, with or without wiper blades, for electric windscreen wipers. However, the subheading does not include lamp holders (subheading 8536 61 10 or 8536 61 90).
8514	Industrial or laboratory electric furnaces and ovens (including those functioning by induction or dielectric loss); other industrial or laboratory equipment for the heat treatment of materials by induction or dielectric loss
8514 20 80	Dielectric furnaces and ovens Microwave ovens of a type designed to be used in restaurants, canteens, etc., differ from domestic appliances of heading 8516 in their power output and oven capacity. Such ovens having a power output of more than 1 000 W or an oven capacity of more than 34 litres are considered to be for industrial use. For microwave ovens combined in a single housing with a grill or another type of oven, the aforementioned power output applies only to the microwave. The classification of such a combination is not influenced by the oven capacity criterion. Microwave ovens having a power output of not more than 1 000 W and an oven capacity of not more than 34 litres are considered to be for domestic use (heading 8516).
8516	Electric instantaneous or storage water heaters and immersion heaters; electric space-heating apparatus and soil-heating apparatus; electrothermic hairdressing apparatus (for example, hairdryers, hair curlers, curling tong heaters) and hand dryers; electric smoothing irons; other electrothermic appliances of a kind used for domestic purposes; electric heating resistors, other than those of heading 8545
8516 10 11 and 8516 10 80	Electric instantaneous or storage water heaters and immersion heaters These subheadings include: 1. water heaters operating both as instantaneous water heaters and water storage heaters; 2. electric boilers which produce only hot water, or hot water and low-pressure steam simultaneously. These subheadings do not include electrically heated steam boilers and 'superheated water' boilers (heading 8402) and electrically heated electric boilers for central heating (heading 8403).

8516 21 00 to 8516 29 99	Electric space-heating apparatus and electric soil-heating apparatus These subheadings include: 1. electric heating equipment for saunas; 2. battery-operated heating devices for warming and defrosting iced-up door locks on motor vehicles by the presence of a heating element in the lock. These small portable devices may incorporate a lamp of the type falling in heading 8513 to illuminate the working area.
8516 50 00	Microwave ovens See the explanatory note to subheading 8514 20 80.
8516 60 10	Cookers (incorporating at least an oven and a hob) Cookers consist of at least a hob and an oven (which may also include a microwave unit or grill).
8516 79 70	Other In addition to the appliances described in the HS Explanatory Note to heading 8516, part (E), (5) to (20), this subheading includes: 1. saunas with infra-red radiators (individual cubicles); 2. foot-heating plates; 3. electrically heated shoe trees; 4. devices for cleaning contact lenses, consisting of two small electrically heated compartments with screw-on lids to hold the contact lenses and heat up the cleansing fluid.
8516 80 20 and 8516 80 80	Electric heating resistors These subheadings include wire, cables, tapes and the like, which are insulated and provide heat for heating ceilings, walls, pipes, containers, etc. However, these subheadings do not include heating resistors connected to parts of equipment, for example, base plates for smoothing irons and plates for electric cookers (subheading 8516 90 00).
8517	Telephone sets, including telephones for cellular networks or for other wireless networks; other apparatus for the transmission or reception of voice, images or other data, including apparatus for communication in a wired or wireless network (such as a local- or wide-area network), other than transmission or reception apparatus of heading 8443, 8525, 8527 or 8528
8517 12 00	Telephones for cellular networks or for other wireless networks See the HS Explanatory Note to heading 8517, part (I), (B). This subheading includes telephones for cellular networks, so called 'mobile phones'. 'Mobile phones' have the following characteristics: — they are 'pocket-sized', i.e. they are of dimensions that do not exceed 170 mm × 100 mm × 45 mm, when measured in their most compact form, — they are capable of operating without an external source of electric power, — they have both a microphone, and an earphone and/or a loudspeaker, either in the same unit or in the form of a detachable headset presented together with the 'mobile phone', for the transmission and reception of voice enabling voice communication, — they incorporate other components such as an amplifier and an antenna for telephony, which provide for dual-way short-range transmission of voice within a network consisting of base stations of subheading 8517 61 and using mobile telephony frequency bands,

- they are able to do telephony communication using cellular networks when equipped with a SIM (Subscriber Identity Module) of various types (physical or software) that has been activated. They may provide that emergency calls may be made without the SIM.

'Mobile phones' may also have other features such as sending and receiving SMS (Short Message Service) and MMS (Multimedia Messaging Service) messages; emails; packet switching for access to the Internet; sending and receiving positioning signals; navigating, routing, maps, instant messaging, VOIP (voice over Internet Protocol); PDA (Personal Digital Assistant); gaming; receiving radio or television signals; capturing, recording and reproducing sound and images.

Irrespective of such additional features, the mobile telephony function is generally the principal function of mobile phones that meet all of the above listed characteristics. This is the case, for example, when the telephony function takes precedence over all other functions, including when incoming calls are normally notified to the user regardless of secondary functions used.

8517 18 00**Other**

In addition to the telephone sets described in the HS Explanatory Note to heading 8517, part (I), (A), this subheading includes push-button, telephone sets incorporating a magnetic card reader, a data display, an electronic circuit-board with microprocessor, several data stores, a clock and a modulator-demodulator (modem). These appliances can be used not only as telephone sets but also (for example, in supermarkets) as data terminals, for example, for verifying magnetic credit cards or cheques or for the transmission of sales data over telephone lines to automatic data-processing machines.

8517 62 00**Machines for the reception, conversion and transmission or regeneration of voice, images or other data, including switching and routing apparatus**

This subheading covers two groups of machines:

1. machines for the reception, conversion and transmission of voice, images or other data;
2. machines for the regeneration of voice, images or other data.

These machines include:

1. network interface cards;
2. modems;
3. repeaters;
4. hubs;
5. bridges (including switches);
6. routers.

This subheading includes apparatus consisting of all the elements required for transmission and reception, in one cabinet or housing. Such is the case, for example, with walkie-talkies, which contain the batteries or accumulators required for their operation, or with transmitter-receiver apparatus, the power supply unit of which would be separate and connected to the apparatus by cable only.

This subheading also includes units in which the transmitter and receiver elements are located in different furniture units or housings, provided that they constitute a functional unit. In order to be regarded as constituting a functional unit, transmitter-receiver apparatus must be installed near each other (for example, in the same building or on the same vehicle) and have certain elements in common, such as the aerial.

This subheading also includes so-called 'wireless headsets' designed to be used solely or principally with telephones for cellular networks (mobile phones), intended to be hung on the ear to provide a hands-free option. They enable the user to control telephone functions, such as answering, ending and dialling (for example, redialling the last call), within approximately 10 m from the mobile phone and have means for adjusting the earphone sound volume. These headsets incorporate a radio transmitter/receiver to communicate with a mobile phone by means of wireless technology, for example, 'Bluetooth'

8517 69 10**Videophones**

See the HS Explanatory Note to heading 8517, part (II), (C).

This subheading also includes closed-circuit television systems consisting of one camera, a panel with several call buttons, and one or more monitors connected to a communication post, and coaxial cables for linking the different elements, presented in the form of a set put up for retail sale.

Products of this subheading may also be combined with an electrical door opener, a signalling device or a light source.

8517 69 30**Reception apparatus for radio-telephony or radio-telegraphy**

This subheading includes:

1. fixed reception apparatus (including those used mainly in large installations), and special devices such as secrecy devices (for example, spectrum inverters); certain receivers, termed 'diversity receivers', employ multiple receiver technique to overcome fading;
2. radiotelephony receivers for motor vehicles, ships, aircraft, trains, etc.;
3. receivers for radiotelegraphic staff location systems;
4. radio receivers for simultaneous interpretation at multilingual conferences;
5. special receivers for distress signals from ships, aircraft, etc.;
6. telemetric signal receivers;
7. 'facsimile' radiotelegraphic apparatus for receiving, on sensitised paper, copies of documents, newspapers, plans, messages, etc.

8517 69 90**Other**

This subheading includes:

1. fixed transmitters. Certain types, used mainly in large installations, include special devices such as secrecy devices (for example spectrum inverters) and multiplex devices (used for sending more than two messages simultaneously);
2. radiotelephony transmitters for motor vehicles, ships, aircraft, trains, etc.;
3. transmitters used in radiotelegraphic staff location systems;
4. radio transmitters for simultaneous interpretation at multilingual conferences;
5. automatic transmitters for distress signals from ships, aircraft, etc.;
6. 'facsimile' radiotelegraphic apparatus for transmitting copies of documents, newspapers, plans, messages, etc.;
7. transmitters of telemetric signals.

8518**Microphones and stands therefor; loudspeakers, whether or not mounted in their enclosures; headphones and earphones, whether or not combined with a microphone, and sets consisting of a microphone and one or more loudspeakers; audio-frequency electric amplifiers; electric sound amplifier sets**

This heading includes separately presented wireless microphones, headphones, earphones and loudspeakers, whether or not combined.

This heading does not include:

1. separately presented cordless handsets for line telephone sets (see the HS Explanatory Note to subheading 8517 62);
2. separately presented wireless headsets (combination of a headphone and a microphone) with dialling capability (see the explanatory note to subheading 8517 62 00).

8519**Sound recording or reproducing apparatus****8519 81 15****Pocket-size cassette-players**

For the determination of the measurements of such apparatus, subheading note 1 to Chapter 85 applies, and the dimensions of the cabinet are to be taken into account; protuberances, such as control buttons, closing devices and securing clips, are to be ignored.

8519 81 35**Other**

This subheading does not include sound reproducing apparatus, so-called 'MP3 players', using optical media and incorporating devices which can be subsequently activated for the reception of radio-broadcast signals by means of software (heading 8527).

8519 81 45	Other The explanatory note to subheading 8519 81 35 applies, <i>mutatis mutandis</i> .
8519 81 95	Other The explanatory note to subheading 8519 81 35 applies, <i>mutatis mutandis</i> .
8522	Parts and accessories suitable for use solely or principally with the apparatus of heading 8519 or 8521
8522 90 41 and 8522 90 49	Electronic assemblies See the explanatory note to subheading 8443 99 10
8523	Discs, tapes, solid-state non-volatile storage devices, ‘smart cards’ and other media for the recording of sound or of other phenomena, whether or not recorded, including matrices and masters for the production of discs, but excluding products of Chapter 37 If media of this heading are presented together with other articles, the following classification principles should be applied: 1. if the media and the other articles make up a set put up for retail sale under general interpretative rule 3(b), the set should be classified by application of that rule, or 2. if the media and the other articles do not make up a set put up for retail sale under general interpretative rule 3(b), then the media should be classified in the appropriate subheading.
8523 21 00	Cards incorporating a magnetic stripe This subheading also includes printed matter, such as travel tickets and boarding passes, that incorporate one or more magnetic stripes.
8523 29 15	Unrecorded This subheading also covers tapes which have to be cut to size before use.
8523 41 10	Discs for laser reading systems of a recording capacity not exceeding 900 megabytes, other than erasable This subheading covers recordable compact discs (CD-Rs). CD-Rs are usually made of colourless transparent polycarbonate with a thickness of 1,2 mm. The top side is coated with a reflective layer of golden or silver colour and can be printed. The underside is covered with a layer of dye and a protective layer. They have a diameter of 8 or 12 cm, or can have the form of a card. Their recording capacity can be checked with reading/writing machines and corresponding recording software. Although recording on such discs can be done in one or several steps, the recorded information cannot be erased.
8523 41 30	Discs for laser reading systems of a recording capacity exceeding 900 megabytes but not exceeding 18 gigabytes, other than erasable This subheading covers recordable digital versatile discs (DVD-/+Rs). In contrast to the CD-Rs of subheading 8523 41 10, DVD-/+Rs consist of two 0,6 mm-thick polycarbonate layers glued together. This physical difference can be identified by examining the outside edge of the disc. They have a diameter of 8 or 12 cm. Their recording capacity can be checked with reading/writing machines and corresponding recording software. Although recording on such discs can be done in one or several steps, the recorded information cannot be erased.

8523 41 90**Other**

This subheading includes:

1. optical discs on which the recorded information can be erased and new data can be recorded ('rewritable compact discs' (CD-RWs) and 'rewritable digital versatile discs' (DVD-/RWs)). They can be identified by the disc recording software of an automatic data-processing machine;
2. rewritable DVD-RAMs and 'MiniDiscs' which are both housed in cartridges.

8523 52 00**'Smart cards'**

See the HS Explanatory Note to heading 8523, (C), (2).

This subheading also includes electronic proximity cards/tags usually consisting of a coil which is activated by a signal from a reader and produces a voltage to power a microcircuit, a code generator which on receiving a signal from the coil generates data, and a signal transmission antenna.

8525**Transmission apparatus for radio-broadcasting or television, whether or not incorporating reception apparatus or sound recording or reproducing apparatus; television cameras, digital cameras and video camera recorders**

This heading includes thermal-imaging cameras with infra-red imagers which capture heat-radiation and convert it into images that represent the temperatures of individual surfaces or objects in different shades of grey or in different colours but which cannot measure temperatures or represent the measured values in figures.

8525 60 00**Transmission apparatus incorporating reception apparatus**

This subheading also includes units in which the transmitter and receiver elements are located in different furniture units or housings, provided that they constitute a functional unit. In order to be regarded as constituting a functional unit, transmitter-receiver apparatus must be installed near each other (for example, in the same building or on the same vehicle) and have certain elements in common, such as the aerial.

**8525 80 11
to
8525 80 99****Television cameras, digital cameras and video camera recorders**

See the HS Explanatory Note to heading 8525, (B).

These subheadings do not cover electronic reading devices for the visually handicapped (see the explanatory note to subheading 8543 70 90).

8525 80 30**Digital cameras**

Digital cameras of this subheading are always capable of still-image recording, whether on an internal storage medium or on interchangeable media. Most of them have the design of a traditional photographic camera.

Digital cameras that are only capable of recording still images remain classified in this subheading.

Cameras of this subheading may also have video-capture capability to record continuous periods of video.

However, when such apparatus are capable, using the maximum storage capacity, of capturing video in a quality of 800 x 600 pixels (or higher) at 23 frames per second (or higher) for a continuous period of at least 30 minutes (regardless of the fact that the captured video images may be recorded in separate files of a duration of less than 30 minutes) they are always to be classified in subheadings 8525 80 91 or 8525 80 99.

If one or more of the above criteria is not fulfilled, the apparatus is to be classified by application of note 3 to Section XVI (see also Commission Implementing Regulation (EU) No 458/2014).

Compared to the video camera recorders of subheadings 8525 80 91 and 8525 80 99, many digital cameras (when functioning as video cameras) do not offer a zoom function during video recording. Unaffected by the storage capacity, some cameras automatically terminate the recording of video after a certain period of time.

8525 80 91
and
8525 80 99

Video camera recorders

Video camera recorders of these subheadings are always capable of recording continuous periods of video, whether on an internal storage medium or on interchangeable media.

In general, the digital video camera recorders of these subheadings have a design which differs from digital cameras of subheading 8525 80 30. They often have a foldable viewfinder and are frequently presented together with a remote control. Most video camera recorders offer a zoom function during video recording. However, the fact that video cameras integrated, for example, into sports goggles have no optical zoom function does not prevent the classification of such products under subheadings 8525 80 91 and 8525 80 99 (see Case C-178/14, *Vario Tek*, ECLI:EU:C:2015:152, para 17 - 29).

Digital video camera recorders that are only capable of recording video are always classified in these subheadings.

Apparatus with both video recording and still image recording capability are to be classified by reference to the explanatory note to subheading 8525 80 30.

These subheadings include remotely controlled apparatus for capturing and recording video and still images which are specifically designed to be used with multi-rotor helicopters (so-called drones), for example, through dedicated contact elements. These apparatus are used for capturing video and aerial still images of the environment and allow the user to visually control the flight of the drone. Such apparatus are always classified under these subheadings regardless of the length of the video recording as the video recording is the principal function. See also the HS classification opinion 8525 80/3.

8525 80 99

Other

This subheading covers video camera recorders (so-called 'camcorders') for the recording not only of sound and images taken by the camera but also of signals from external sources, for example, DVD-players, automatic data-processing machines or television receivers. The images thus recorded can be reproduced by means of an external television receiver or monitor.

This subheading also includes 'camcorders' in which the video input is obstructed by a plate or in another way, or in which the video interface can be subsequently activated as video input by means of software. The apparatus is nevertheless designed to record television programmes or other externally incoming signals.

However, 'camcorders' with which only the images taken by the camera can be recorded and reproduced by means of an external television receiver or monitor fall in subheading 8525 80 91.

8527

Reception apparatus for radio broadcasting, whether or not combined, in the same housing, with sound recording or reproducing apparatus or a clock

For transmission apparatus incorporating reception apparatus, see the explanatory note to subheading 8525 60 00.

8527 12 10
and
8527 12 90

Pocket-size radio cassette players

See the explanatory note to subheading 8519 81 15.

8527 13 10

With laser reading system

This subheading includes radio-broadcast receivers, combined with a sound recording or reproducing apparatus with a laser reading system, whether or not combined with other sound recording or reproducing apparatus (for example of the cassette-type) or a clock.

This subheading also includes sound reproducing apparatus, so-called 'MP3 players', using optical media and incorporating devices which can be subsequently activated for the reception of radio-broadcast signals by means of software.

8527 13 99

Other

The explanatory note to subheading 8527 13 10 applies, *mutatis mutandis*.

8527 21 20

With laser reading system

The explanatory note to subheading 8527 13 10 applies, *mutatis mutandis*.

8527 21 70

With laser reading system

The explanatory note to subheading 8527 13 10 applies, *mutatis mutandis*.

8527 91 11 to 8527 91 99	Combined with sound recording or reproducing apparatus Stereo systems (hi-fi systems) containing a radio receiver, put up in sets for retail sale, consisting of modular units in their own separate housings, for example, in combination with a CD player, a cassette recorder, an amplifier with equalizer, always fall in these subheadings as the radio receiver gives them their essential character.
8527 91 11 and 8527 91 19	Within the same housing one or more loudspeakers These subheadings cover apparatus from which the loudspeakers are inseparable. However, these subheadings do not cover apparatus with detachable loudspeakers, even if the loudspeakers can be attached with fastening devices (subheading 8527 91 35, 8527 91 91 or 8527 91 99).
8527 91 35 to 8527 91 99	Other If loudspeakers for stereo systems (hi-fi systems) are specially designed for and packed together with other components of the sets, they are included in these subheadings.
8527 91 35	With laser reading system The explanatory note to subheading 8527 13 10 applies, <i>mutatis mutandis</i>.
8528	Monitors and projectors, not incorporating television reception apparatus; reception apparatus for television, whether or not incorporating radio-broadcast receivers or sound or video recording or reproducing apparatus According to general rule 3(c) for the interpretation of the Combined Nomenclature, video surveillance systems, consisting of a limited number of television cameras and one monitor, fall within this heading when put up in a set for retail sale. The components of such video surveillance systems are to be classified separately when the system is not put up in a set for retail sale (see the HS General Explanatory Note to Section XVI, (VII)). This heading does not cover electronic reading devices for the visually handicapped (see the explanatory note to subheading 8543 70 90).
8528 62 00 to 8528 69 80	Projectors These subheadings do not include products comprising a projector and a screen in the same housing (subheadings 8528 52 10 to 8528 59 00 or, when incorporating reception apparatus for television, subheading 8528 72 10).
8528 71 11 to 8528 71 19	Video tuners These subheadings include apparatus incorporating a video tuner which converts high-frequency television signals into signals usable by video recording or reproducing apparatus or monitors. The apparatus incorporates selector circuits, which permit tuning to a special channel or carrier frequency, and demodulation circuits. The apparatus may also be equipped with a decoding device (colour) or with separator circuitry for synchronisation. It is generally designed to operate with an individual aerial or a shared aerial (high-frequency cable distribution). The output signal can be used as an input signal for monitors or for recording or reproducing apparatus. It consists of the original camera signal (i.e., not modulated for transmission purposes). Analogue video tuners within the meaning of these subheadings may be in the form of modules comprising, at least, the radio frequency circuits (RF block), intermediate frequency circuits (IF block) and demodulation circuits (DEM block) where the output is a separate audio and composite video baseband signal (CVBS). Digital video tuners within the meaning of these subheadings may be in the form of modules comprising, at least, the RF block, IF block, DEM block and an MPEG decoder for digital television where the output is a separate audio and digital video signal. Modules incorporating both an analogue and a digital video tuner component fall under these subheadings when one of the components is to be classified as a complete or finished video tuner by application of general rule 2(a) for the interpretation of the Combined Nomenclature. A module not fulfilling the above conditions is to be classified as a 'part' under heading 8529.

8528 71 11

Electronic assemblies for incorporation into automatic data-processing machines

For the definition of the expression 'electronic assemblies', see the explanatory note to subheading 8443 99 10.

8528 72 10

Television projection equipment

This subheading includes projection equipment incorporating reception apparatus for television. Such equipment projects the image via an optical system onto a screen. It may be based on cathode-ray-tube or flat-panel (for example, DMD, LCD, plasma) technology.

The projection screen may be either incorporated in the same housing as the television receiver or may be separate.

8528 72 30

With integral tube

This subheading includes apparatus combining tuner and monitor functions in the same housing, with possible simultaneous use of certain elements of each. Domestic television receivers generally belong to this category.

The diagonal measurement of the screen means the active part of the picture tube measured in a straight line.

8529

Parts suitable for use solely or principally with the apparatus of headings 8525 to 85288529 90 20
to
8529 90 97**Other**

These subheadings include:

1. dials;
2. tuning blocks;
3. diaphragms for television cameras;
4. so-called PAL-Secam adapters. These are decoder boards (printed circuit boards mounted with electrical components) for subsequent incorporation in television receivers designed for the reception of PAL signals, so as to make these sets suitable for the dual reception of PAL/Secam signals.

These subheadings do not include waveguides (classification of the tubes according to their constituent material).

8529 90 41
and
8529 90 49**Cabinets and cases**

The terms 'cabinets' and 'cases' include housings not equipped with electric or electronic components (for example, printed circuit boards, loudspeakers, cables, etc.).

'Cabinets and cases' equipped with electric or electronic components are excluded (for example, subheading 8529 90 92 or 8529 90 97).

Parts of 'cabinets and cases' are excluded from these subheadings.

8529 90 65

Electronic assemblies

See the explanatory note to subheading 8443 99 10.

8529 90 92

For television cameras of subheadings 8525 80 11 and 8525 80 19 and apparatus of headings 8527 and 8528

This subheading includes so-called 'LCD modules' consisting of a thin-film transistor (TFT) liquid-crystal layer sandwiched between two sheets or plates of glass or plastic, not combined with touch-screen facilities, used in the manufacture of monitors and/or reception apparatus for television of heading 8528.

The modules are equipped with one or more printed or integrated circuits, with control electronics for pixel-addressing only. They are not equipped with electronics (such as a video converter or a scaler) for video processing.

The modules can be equipped with a backlight unit and/or inverters.

This subheading does not include liquid-crystal devices consisting of a liquid-crystal layer sandwiched between two sheets or plates of glass or plastics, whether or not fitted with electrical connections (i.e. for the supply of power), presented in the piece or cut to special shapes (subheading 9013 80 20).

8531 Electric sound or visual signalling apparatus (for example, bells, sirens, indicator panels, burglar or fire alarms), other than those of heading 8512 or 8530

8531 10 30 and 8531 10 95 Burglar and fire alarms and similar apparatus

See the HS Explanatory Note to heading 8531, third paragraph, (E).

These subheadings do not include burglar alarms of a kind used for motor vehicles (subheading 8512 30 10).

8531 20 20 to 8531 20 95 Indicator panels incorporating liquid crystal devices (LCD) or light-emitting diodes (LED)

These subheadings include variable electroluminescent diode readouts used principally as numerical and/or alphanumeric signals or display panels, for example, made up of cells or various light-emitting diodes. Each sign contains a certain number of light-emitting diodes in the form of components which are either discrete or attached to a single microchip. These devices are mounted on a printed circuit with a driver/decoder. Each digit or assembly of digits is covered by translucent material which amplifies the intensity of the light points produced by the diodes in order to produce figures or letters as a function of the pulse applied to the circuit by an input signal.

8531 90 00 Parts

This subheading includes anti-theft tags (such as those consisting of a diode and an antenna, so-called 'microwave tags', or those consisting of a microchip-resonant circuit, so-called 'frequency division' tags) attached to goods which, when they come within the signal field of a burglar alarm system installed at a shop exit, of a type falling in subheading 8531 10, influence the receiving electronics of the system and trigger off an alarm.

This subheading also includes tags which combine two different technologies, such as acousto-magnetic with microwave or acousto-magnetic with radio frequency.

However, this subheading does not include:

- tags consisting of a wire, strip or combination of two strips, so-called 'magnetic tags' or 'acousto-magnetic tags', of magnetic materials intended to be used within the control zone of an Electronic Article Surveillance (EAS) system (heading 8505),
- tags which have embedded in them one or more electronic integrated circuits in the form of chips and an antenna but with no other active or passive circuit elements ('smart cards'), so-called 'Radio Frequency Identification' tags (RFID) (subheading 8523 52),
- tags containing one or more electronic integrated circuits in the form of chips with an embedded antenna and one or more capacitors but no other active or passive circuit elements (subheading 8523 59),
- tags in the form of a printed circuit (see note 6 to Chapter 85), so-called 'radio frequency tags' (heading 8534 00).

8534 00 Printed circuits

This heading includes:

1. paper tags in the form of a printed circuit, to prevent the theft of, for example, library books;
2. flexible electrical conductors in the form of printed circuits.

8536 Electrical apparatus for switching or protecting electrical circuits, or for making connections to or in electrical circuits (for example, switches, relays, fuses, surge suppressors, plugs, sockets, lamp holders and other connectors, junction boxes), for a voltage not exceeding 1 000 V; connectors for optical fibres, optical-fibre bundles or cables

In addition to the apparatus described in the HS Explanatory Note to heading 8536, this heading includes:

1. distribution frames for telephone exchanges;
2. intersection or junction plates for overhead tram cables;
3. junction plates for electric motors;
4. commutators for multiple point transformers;
5. connecting parts for accumulator elements;

6. so-called inductive proximity switches, i.e., electronic switches incorporating a free-radiating induction coil for the contactless actuation (closure) of the switch when a metal object comes within the stray field of the induction coil. Such switches are used, for example, in machine tools, conveyor machinery and weighing balances instead of mechanically operated electrical terminal switches;
7. on-off switches housed alongside an intensity modulation device (dimmer) in a single housing. By means of this equipment, it is possible not only to switch mains-operated lamps on and off, but also to vary their intensity continuously;
8. so-called contact pads for connecting electrical circuits. These consist of two superposed resilient films of plastics in which are embedded, at regular intervals, a large number of electrically conductive contact points of silicone rubber. These contact pads are incorporated, for example, under the keys of telephone sets. Each time a key is pressed the appropriate contact points on the two films touch, and electrical contact is established;
9. electronic switches which operate by contactless means, using semiconductor components (for example, transistors, thyristors, integrated circuits).

However, this heading does not include:

- (a) insulation and sheathing for electric lines (generally, heading 7326);
- (b) control equipment for railways and other communication routes (heading 8530).

8536 50 11

Push-button switches

This subheading does not cover touch-sensitive switches (subheading 8536 50 19).

8536 69 10 to 8536 69 90

Other

These subheadings include electromechanical plugs (so-called male type connectors) and sockets (so-called female type connectors) which permit connection, for example, between appliances, cables and connector boards.

The connectors may have a plug or socket on both sides or a plug or socket on one side and another contact device on the other (for example, crimp, clamp, soldered or screw-type) see the examples in figures 1 and 2.

Figure 1:

On one side the cable is connected to the connector by means of a crimp/clamp connection. On the other side the connection is made by plugging the male-type/female type connector.



Figure 2:

On one side, the cable is connected to the connector by means of a crimp/clamp connection. On the other side, the connection is made by plugging the female-type connector. In addition, this connector is also fitted with an additional connection in the form of a male-type connector.

These subheadings also include plug-in couplings comprising a plug and socket pair (two pieces). The plug and the socket pieces each have one socket and one other contact device.

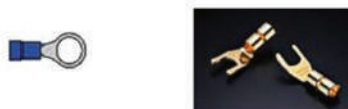


These subheadings do not include connection or contact elements with which an electrical connection is solely established by other means (for example, crimp, screw-type, soldered or clamp terminals). These are covered by subheading 8536 90 (see the examples in figures 3 to 7).

However plugs and sockets, threaded internally or externally, are included.

Figures 3 and 4:

On one side the cable is connected to the connector by means of a crimp/clamp connection. On the other side the connection is made by using a screw or a bolt and nut.



Figures 5, 6 and 7:

The electrical connection is not made by plugging. There is no male or female type connector present. The electrical connection is made by screwing or clamping.



8536 69 10

For coaxial cables

This subheading covers only plugs and sockets used for coaxial connections (see the examples in figures 1 to 6).

Figure 1
Coaxial plug

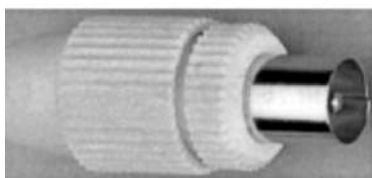


Figure 2
Coaxial socket



Figure 3
Coaxial plug

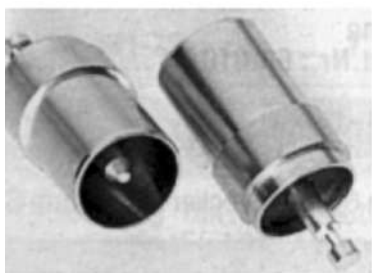


Figure 4
Coaxial socket

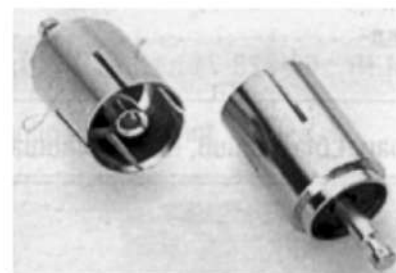


Figure 5

Threaded coaxial plug

*Figure 6*

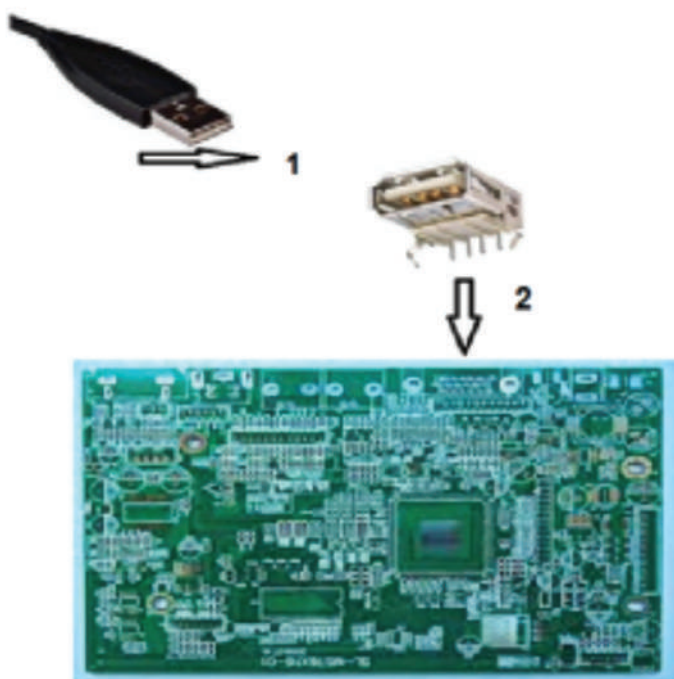
Coaxial adapter



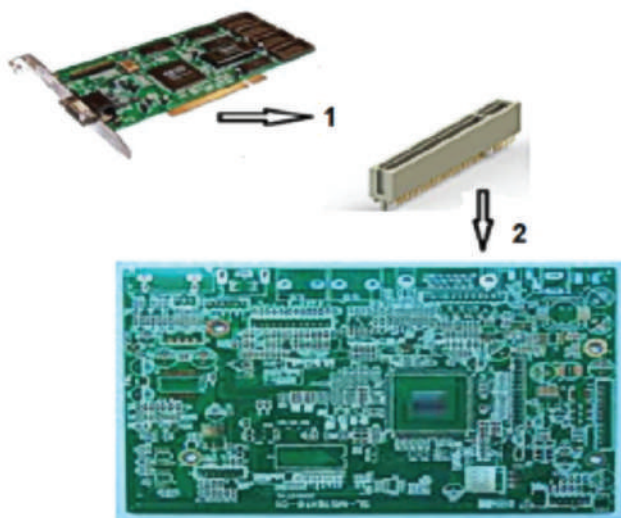
8536 69 30

For printed circuits

This subheading covers plugs and sockets that are intended to be directly attached to a printed circuit board (PCB) or flexible printed circuit on at least one side of the plug or socket (see the examples below).



1. This USB connector is not intended to be directly attached to a PCB and is therefore excluded from this subheading.
2. This USB receptacle is intended to be directly attached to a PCB and is therefore included within this subheading.



1. This connector is not intended to be directly attached to a PCB and is therefore excluded from this subheading.
2. This connector is intended to be directly attached to a PCB and is therefore included within this subheading.

8536 70 00**Connectors for optical fibres, optical-fibre bundles or cables**

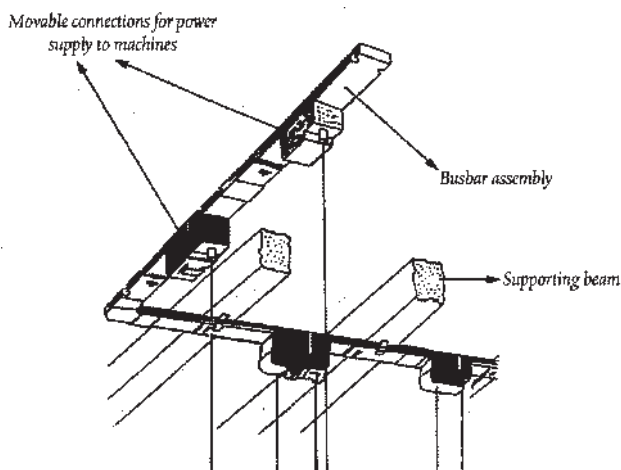
See note 6 to this chapter.

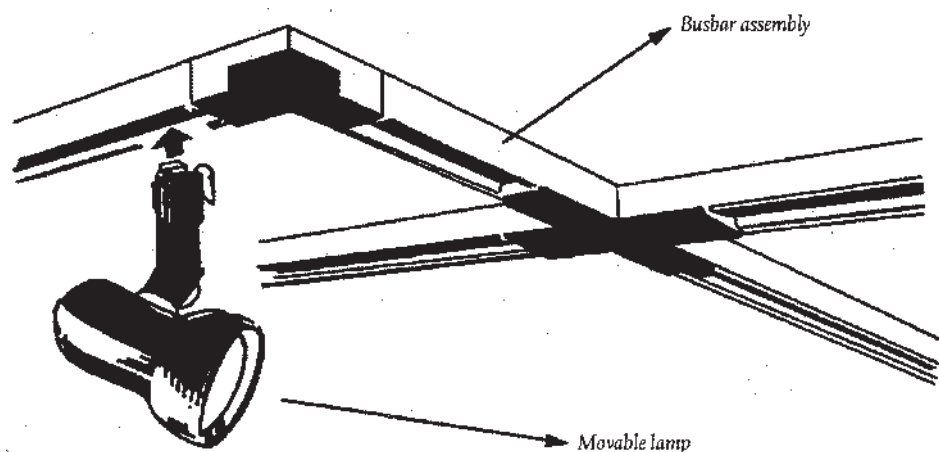
See the HS Explanatory Note to heading 8536, (IV).

8536 90 01**Prefabricated elements for electrical circuits**

This subheading includes ready-to-mount elements for the transmission of electricity. Such elements provide a degree of spatial flexibility in the power supply for lamps and electrically powered machines and equipment. The contact points across which the current is delivered are of the clamp or sliding variety.

Typical applications are shown in the diagrams below:



**8536 90 10****Connections and contact elements for wires and cables**

This subheading covers all terminal devices fitted to the ends of wires or cables for establishing an electrical connection by means other than plugging (for example, crimp, screw-type, soldered or clamp terminals).

8537

Boards, panels, consoles, desks, cabinets and other bases, equipped with two or more apparatus of heading 8535 or 8536, for electric control or the distribution of electricity, including those incorporating instruments or apparatus of Chapter 90, and numerical control apparatus, other than switching apparatus of heading 8517

This heading includes assemblies on a support (for example, panel, box) of identical items of equipment to those of heading 8536 (for example, light switches).

8537 10 91**Programmable memory controllers**

This subheading includes panel cabinets incorporating application memory for the electrical control of machines. They are fitted not only with equipment of heading 8535 or 8536 (for example, relays) but also, for example, with transistors or triacs of heading 8541 as switching devices and which, in addition to these switching elements, also include microprocessors (for example, for logic processing and the management of inputs and outputs), interfaces and a power supply unit (power pack).

8537 10 98**Other**

This subheading includes:

1. remote cable control units for video recorders. These are electrical control pads (with push-button switches and other electrical components mounted on a board) which are used to execute, via a connecting cable, all the functions of the linked-up video recorder;
2. electrical control devices for automatic vending machines. These consist of a control pad incorporating, in addition to relays and electrical switches, triacs and integrated circuits.

This subheading does not include:

- (a) electrical control devices known as 'joysticks' within the meaning of note 5(C) to Chapter 84 (heading 8471);
- (b) cordless infra-red devices for the remote control of television receivers, video recorders or other electrical equipment (heading 8543).

8538**Parts suitable for use solely or principally with the apparatus of heading 8535, 8536 or 8537****8538 90 11****Electronic assemblies**

See the explanatory note to subheading 8443 99 10.

8538 90 91**Electronic assemblies**

See the explanatory note to subheading 8443 99 10.

8539**Electric filament or discharge lamps, including sealed beam lamp units and ultraviolet or infra-red lamps; arc lamps; light-emitting diode (LED) lamps**

With the exception of certain arc lamps (see the explanatory note to subheadings 8539 41 00 and 8539 49 00) which are specially equipped or mounted, only lamps and tubes as such and parts thereof which are identifiable in accordance with note 2(b) to section XVI, fall in this heading.

Apparatus fitted with such lamps (consisting, for example, of a simple reflector with a support or base) is classified in its appropriate heading, as lighting equipment (heading 9405), heating equipment (heading 7321, for example) or medical equipment (heading 9018), etc.

**8539 21 30
to
8539 29 98****Other filament lamps, excluding ultraviolet or infra-red lamps**

These subheadings do not include:

- (a) resistance lamps with carbon filaments and variable resistance lamps with iron filaments in hydrogen (heading 8533);
- (b) electric lamp sets of a specific length comprising a fixed number of fancy lights, mainly used for decorating Christmas trees (heading 9405).

**8539 31 10
to
8539 39 80****Discharge lamps, other than ultraviolet lamps**

These subheadings include:

- 1. xenon tubes;
- 2. spectral discharge lamps;
- 3. glow discharge lamps;
- 4. alphanumeric tubes.

**8539 41 00
and
8539 49 00****Ultraviolet or infra-red lamps; arc lamps**

These subheadings include:

- 1. ultraviolet tubes and lamps. In addition to the uses referred to in the HS Explanatory Note to heading 8539, part (D), first paragraph, the following may be mentioned:
 - (a) actino-chemical printing;
 - (b) vitaminisation;
 - (c) sterilisation;
 - (d) photochemistry;
 - (e) production of ozone.
- 2. infra-red lamps and tubes. In addition to the uses referred to in the HS Explanatory Note to heading 8539, part (D), second paragraph, the following may be mentioned:
 - (a) space heating;
 - (b) equipment for infra-red emitters (for example, in anti-theft apparatus);
 - (c) scientific research (for example, spectral analysis).
- 3. arc lamps described in the HS Explanatory Note to heading 8539, part (E).

Carbon-electrode arc lamps are installed, for example, in cinematographic projection equipment of heading 9007 or used in document-reproducing equipment.

Subheading 8539 41 00 includes electric arc lamps equipped with special supports as well as guidable light equipment consisting of one or more arc lamps mounted on a mobile support, intended for use particularly in photographic or cinematographic studios.

These subheadings do not include:

- (a) luminescent diodes (heading 8541);
- (b) electroluminescent devices, generally in strips, plates or panels, and based on electroluminescent substances (for example, zinc sulphide) placed between two layers of conductive material (heading 8543).

**8539 90 10
and
8539 90 90**

Parts

These subheadings include:

- 1. bases for incandescent and discharge electric lamps and bulbs;
- 2. tungsten filaments, spiral, cut to length and ready for mounting;
- 3. metal electrodes for discharge lamps and tubes;
- 4. glass parts (excluding insulating parts of subheading 8547 90 00) intended for mounting inside lamps or tubes;
- 5. supporting brackets for filaments.

8540

Thermionic, cold cathode or photocathode valves and tubes (for example, vacuum or vapour or gas-filled valves and tubes, mercury arc rectifying valves and tubes, cathode ray tubes, television camera tubes)

8540 11 00

Colour

This subheading covers the cathode ray tubes described in the HS Explanatory Note to heading 8540, fourth paragraph, (2), and which comply with the conditions set out below:

- 1. slit mask;
- 2. gap between two lines of the same colour at the centre of the screen not less than 0,4 mm.

For the diagonal measurement of the screen, see the explanatory note to subheading 8528 72 30.

8540 60 00

Other cathode ray tubes

This subheading covers the cathode ray tubes described in the HS Explanatory Note to heading 8540, fourth paragraph, (2), (d), other than those covered by subheadings 8540 11 and 8540 12.

**8540 71 00
and
8540 79 00**

Microwave tubes (for example, magnetrons, klystrons, travelling wave tubes, carcinotrons), excluding grid-controlled tubes

These subheadings cover the tubes described in the HS Explanatory Note to heading 8540, fourth paragraph, (4).

These subheadings do not include Geiger-Müller counter tubes (subheading 9030 90 00).

**8540 81 00
and
8540 89 00**

Other valves and tubes

In addition to the lamps, tubes and valves for rectifying electric current referred to in the HS Explanatory Note to heading 8540, fourth paragraph, (1), these subheadings include phanotrons, thyratrons, ignitrons and high-voltage rectifying tubes for X-ray apparatus.

8541

Diodes, transistors and similar semiconductor devices; photosensitive semiconductor devices, including photovoltaic cells whether or not assembled in modules or made up into panels; light-emitting diodes (LED); mounted piezoelectric crystals

8541 40 90

Other

This subheading includes photovoltaic cells assembled in modules or made up into panels incorporating bypass diodes (but no blocking diodes). The bypass diodes are not elements which supply the power directly to, for example, a motor (see the HS Explanatory Notes to headings 8501 and 8541).

8541 90 00**Parts**

In addition to the parts mentioned in the HS Explanatory Note to heading 8541, this subheading includes:

1. supports and housings for piezoelectric crystals;
2. housings of metal, steatite, etc., for mounted semiconductors.

This subheading does not include:

- (a) electrical connectors for conducting current between pins (or contacts) and electrodes (heading 8536);
- (b) graphite parts (subheading 8545 90 90).

8542**Electronic integrated circuits**

This heading includes interchangeable pre-programmed memory modules in the form of a monolithic integrated circuit for electronic translation devices falling in subheadings 8470 10 00 and 8543 70 10.

This heading does not include discs (sometimes referred to as wafers) which, for the purposes of their applications in the field of electronics, consist of doped chemical elements, whether or not polished and whether or not epitaxially coated, provided that they have not been selectively doped or diffused to create discrete regions (heading 3818 00).

**8542 31 11
to
8542 31 90****Processors and controllers, whether or not combined with memories, converters, logic circuits, amplifiers, clock and timing circuits, or other circuits**

These subheadings include:

1. microprocessors, also called micro processing units (MPUs), which are integrated circuits that can be defined as devices that perform the primary instruction, execution and system control functions. They consist of the following main parts:
 - the arithmetic logic unit (ALU);
 - the instruction decoder and programme counter;
 - the control unit;
 - the input/output (I/O) unit, for communications with other devices.

A microprocessor can operate only if, apart from the internal memory, an external memory or other device is being used.

They may have one or more micro-program-memories (RAM or ROM) for loading or storing micro-instructions, thereby increasing the number of primary instructions in the control unit.

The micro-programmed ROM that a microprocessor may contain is dedicated to storing binary basic-instructions and is not to be considered as a real program memory storing the instructions to be executed.

These products may contain an instruction cache memory or micro-peripheral functions.

Microprocessors entirely processed for a specific application, based on 'full custom', 'gate array' or 'standard cell' designs, are included.

The processing capacity of microprocessors means the length of each word which the accumulator of the arithmetic and logic unit can process in a single micro-instruction cycle;

2. microcontrollers and microcomputers which are integrated circuits, consisting of at least the following main parts:
 - a microprocessor, also called micro processing unit (MPU);
 - a program memory (for example, RAM, ROM, PROM, EPROM, E²ROM, flash E²PROM) linked to the instruction decoder and containing a program defining an instruction sequence;
 - a data memory (for example, RAM or E²PROM), not accessible from outside the chip contrary to microprocessors;
 - an external bus (for data, addresses or instructions).

Microcontrollers are programmed, or further programmable, to perform specific functions and only for certain use (for example, for television receivers, video recording or reproducing apparatus or microwave ovens).

Microcomputers are capable of operating independently (stand alone) and for general use (for example, mainframes, minicomputers and personal computers). Microcomputers are freely programmable in accordance with the requirements of the user.

Microcontrollers, entirely processed for a specific application, based on 'full custom', 'gate array' or 'standard cell' designs, are included.

The processing capacity of microcontrollers or microcomputers means the length of each word which the accumulator of the arithmetic logic unit can process in a single micro-instruction cycle;

3. control circuits which are integrated circuits used to influence a process or the performance of a machine (for example, an automatic data-processing machine). Control circuits (for example, for disk storage units, memories, electric motors or cathode ray tubes) are generally capable of interpreting signals and, in accordance with the interpretation, determining the timing and sequence of certain functions (for example, input, processing, storage and output in an automatic data-processing system).

**8542 31 11
and
8542 31 19**

Goods specified in note 9(b)(3 and 4) to this chapter

See the HS Explanatory Note to heading 8542, (III).

**8542 32 11
and
8542 32 19**

Goods specified in note 9(b)(3 and 4) to this chapter

See the HS Explanatory Note to heading 8542, (III).

8542 32 45

Static random-access memories (S-RAMs), including cache random-access memories (cache-RAMs)

Cache random-access memories (cache-RAMs) are static random-access memories with an access time faster than the main memory. Cache-RAMs are generally used as temporary buffer memories to accommodate the speed differential between a central processing unit and the main memory.

**8542 32 61
to
8542 32 75**

Electrically erasable, programmable, read only memories (E²PROMs), including flash E²PROMs

E²PROMs are in general byte-erasable memories.

Flash E²PROMs are memories also called 'flash Memories', 'flash EPROMs' or 'flash EEPROMs'.

Flash memories may be based on EPROM or E²PROM-technology and are electrically erasable, either totally (bulk-wise) or by sector (page- or block-wise).

Programming, reading and erasure of these memories can be accomplished at a dual power supply or at a single power supply.

Flash memories based on EPROM technology have an array structure composed of one transistor cell.

Flash memories based on E²PROM technology have an array structure composed of two or more transistor cells or of one transistor cell combined with another transistor per sector (page or block). The latter type further differentiates from EPROM-technology based memories because they contain a number of characteristic E²PROM elements (for example, an E²PROM command-set).

8542 32 90

Other memories

This subheading includes content addressable memories (CAMs) and ferroelectric memories.

Content addressable memories (CAMs) are content associative storage devices. Storage locations of these devices are identified by their contents or by part of their contents, rather than by their names or positions (addresses).

Ferroelectric memories are non-volatile memories obtained by combining ferroelectric and semiconductor material. The ferroelectric material is able to retain electric polarisation in the absence of an applied electric field.

These devices are both electrically programmable and erasable.

8542 39 11
to
8542 39 90

Other

These subheadings include:

1. full custom logic circuits which are defined by and produced for a single user. The manufacturing process involves routing and placement of cells (logic gates), using fully customised diffusion masks. Full custom logic circuits are designed to perform application-specific functions. They are known as application-specific integrated circuits, so called ASICs;
2. gate arrays which are integrated logic circuits, consisting of a fixed and regular arrangement of uncommitted logic elements (transistor cells of, for example, AND-, NAND-, OR- or NOR- gates). Gate arrays are programmed in accordance with user-specific interconnection of these logic elements by one or more metallisation patterns;
3. standard cells which are integrated logic circuits, consisting of a user-specific arrangement of predefined subcircuits and fixed subcircuits. These subcircuits may contain any integrated function (for example, a logic function or a memory function);
4. programmable logic circuits which are integrated circuits, consisting of fixed logic elements. The eventual function of these circuits is determined by the user by blowing fusible links or by electrically programming the interconnections between the logic elements;
5. standard logic circuits which are integrated logic circuits consisting of less than 150 logic gates (for example, AND, NAND, OR, NOR). These devices may integrate several functions or sets of identical and independent functions;
6. interface circuits which are integrated circuits which perform a link function (for example, by code conversion, by conversion between bit-serial and bit-parallel or by synchronization) to interconnect programs, integrated circuits, peripherals or systems with different characteristics;
7. microperipherals which are integrated circuits performing specific functions to complement microprocessors, microcontrollers or microcomputers and improve their external communication, control and interface features.

Technical specifications of a microperipheral clearly express its relationship with and dedication to a microprocessor, microcontroller or microcomputer.

Communications, control and interface features may consist of bus controllers, memory controllers (DRAM controllers, memory management units (MMUs), direct memory access controllers) or peripheral interface controllers (graphic controllers, local area network controllers, universal asynchronous receiver/transmitter controllers, keyboard controllers, mass storage controllers);

8. smartpower circuits which are integrated analogue circuits which combine digital and analogue circuitry (power transistors) to control logic output signals and power output signals. These devices are able to provide, for example, internal power dissipation protection, fault management or diagnostic capabilities.

These subheadings do not include programmable read only memories (PROMs) (subheading 8542 32 90).

8542 39 11
and
8542 39 19

Goods specified in note 9(b)(3 and 4) to this chapter

See the HS Explanatory Note to heading 8542, (III).

8543

Electrical machines and apparatus, having individual functions, not specified or included elsewhere in this chapter

8543 70 90

Other

This subheading includes:

1. electrostatic devices (for example, for flocking machines);
2. electroluminescent devices, generally in the form of strips, plates or panels;
3. thermo-electric generators consisting of a thermopile comprising a variable number of thermocouples and a heat source (for example, butane gas), producing direct current by the Seebeck effect;
4. static electricity eliminator devices;
5. demagnetising equipment;

6. shock-wave generators;
7. digital flight-data recorders (flight recorders) in the form of a fire-proof, crash-proof electronic apparatus for the continuous in-flight recording of specific flight data;
8. cordless infra-red devices for the remote control of television receivers, video recorders or other electrical equipment;
9. electronic sound-effect devices used as peripherals for electric guitars for the production of various effects (for example, sound duplication, sound distortion, echoing). These devices are not housed in the guitar casing but are wired between the guitar and the power amplifier;
10. electronic reading devices for the visually handicapped. These devices comprise, in a single housing, a camera, which scans the original text (for example, newspaper or book) and a monitor which reproduces the text in magnified form.

This subheading also covers small electronic devices not mounted on a base-plate (including so-called 'minicomputers'), which can be used to form words and sentences which are translated into a chosen foreign language depending on the memory modules used with these devices. They have an alphanumeric keyboard and a rectangular display. However, this subheading does not cover similar devices with calculating functions (heading 8470).

This subheading does not include:

- (a) electrostatic filters and electromagnetic water purifiers (heading 8421);
- (b) ultraviolet irradiation equipment for treating milk (heading 8434);
- (c) equipment for cleaning various articles ultrasonically (metal parts, in particular) and ultrasonic generators (heading 8479);
- (d) ultrasonic welding equipment (heading 8515);
- (e) ultraviolet irradiation equipment for medical purposes, even if a practitioner is not needed to use them (heading 9018);
- (f) electric regulators for regulating electric or non-electric variables, falling in heading 9032.

8544

Insulated (including enamelled or anodised) wire, cable (including coaxial cable) and other insulated electric conductors, whether or not fitted with connectors; optical-fibre cables, made up of individually sheathed fibres, whether or not assembled with electric conductors or fitted with connectors

This heading does not include:

- (a) flexible electrical conductors, in the form of printed circuits (heading 8534 00);
- (b) separately presented connectors and adapters, for cables or other electrical conductors (heading 8535 or 8536);
- (c) connectors for optical-fibre cables (subheading 8536 70 00).

8544 42 10

Of a kind used for telecommunications

See the HS Explanatory Note to heading 8544.

For the purposes of this subheading, the expression 'of a kind used for telecommunication' includes electrical conductors fitted with connectors used in telecommunication networks, for example, to connect an automatic data-processing (ADP) machine with a modem.

However, this subheading does not include:

- (a) electrical conductors fitted with connectors, to be used for connecting different apparatus (for example, a DVD player with a monitor, or an ADP machine with a monitor, a printer, a keyboard, a projector, etc.);
- (b) electrical conductors designed to be inserted into machines, for example, an ADP machine or a unit thereof, to connect different internal parts of the machine;
- (c) electrical conductors which serve only to supply power (for example, power cables).

8544 49 20

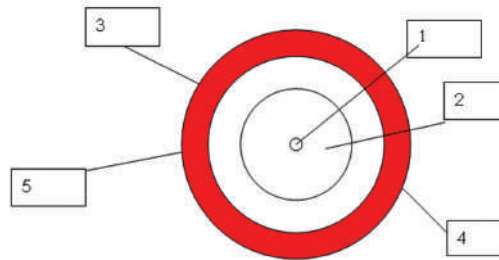
Of a kind used for telecommunications, for a voltage not exceeding 80 V

The explanatory note to subheading 8544 42 10 applies, *mutatis mutandis*.

8544 70 00**Optical-fibre cables**

This subheading also includes optical-fibre cables, designed for example for telecommunication use, made up of optical fibres individually coated with a dual layer of acrylate polymer placed in a protective casing. The coating consists of an inner sheath of soft acrylate and an outer sheath of hard acrylate, the latter being coated by a layer of various colours.

The coating of the individual optical fibres provides protection and structural integrity, for example by protecting the individual fibres against fracture.



1. optical-fibre core (glass-fibre core);
2. optical-fibre cladding (glass);
3. inner sheath of soft acrylate coating;
4. colour coding (ColourLock) identification;
5. outer sheath of hard acrylate coating.

8545**Carbon electrodes, carbon brushes, lamp carbons, battery carbons and other articles of graphite or other carbon, with or without metal, of a kind used for electrical purposes****8545 90 90****Other**

This subheading includes:

1. carbons for lightning arresters;
2. contacts for electrical control equipment or for rheostats.

This subheading does not include compositions of paste for electrodes with a basis of carbonised material (heading 3824).

8547**Insulating fittings for electrical machines, appliances or equipment, being fittings wholly of insulating material apart from any minor components of metal (for example, threaded sockets) incorporated during moulding solely for purposes of assembly, other than insulators of heading 8546; electrical conduit tubing and joints therefor, of base metal lined with insulating material****8547 20 00****Insulating fittings of plastics**

This subheading includes insulating parts obtained by compressing glass fibres steeped in resin or by superimposing or compressing layers of paper or woven materials previously impregnated with artificial resins, provided that the products in question are hard and rigid (see the HS General Explanatory Note to Chapter 39, 'Combinations of plastics and materials other than textiles', (d)).

8547 90 00**Other**

This subheading includes insulating fittings of paper or paperboard, asbestos cement and mica, as well as the conduit tubing and joints therefor referred to in the HS Explanatory Note to heading 8547, (B).

8548 **Waste and scrap of primary cells, primary batteries and electric accumulators; spent primary cells, spent primary batteries and spent electric accumulators; electrical parts of machinery or apparatus, not specified or included elsewhere in this chapter**

8548 90 90 **Other**

This subheading includes:

1. electric coils without magnetic cores which are equally suitable for use in machines or apparatus falling in different headings, for example, for transformers of heading 8504 and for electromagnets of heading 8505 (application of note 2(c) to Section XVI);
2. delay lines used, for example, in automatic data-processing machines or television receivers;
3. electronic filters used for the selective transmission of different frequencies of electromagnetic, electromechanical or electro-acoustic vibrations;
4. elements of ferrite or other ceramics (for example, those used in circulators for ultra-high-frequency transmission apparatus or as a high-frequency filter in electrical cables), which are electrical parts, equally suitable for use in machines or apparatus falling within different headings of this chapter.

SECTION XVII

VEHICLES, AIRCRAFT, VESSELS AND ASSOCIATED TRANSPORT EQUIPMENT

Additional note 2 The explanatory note to additional note 3 to section XVI applies, *mutatis mutandis*.

CHAPTER 86

**RAILWAY OR TRAMWAY LOCOMOTIVES, ROLLING STOCK AND PARTS THEREOF;
RAILWAY OR TRAMWAY TRACK FIXTURES AND FITTINGS AND PARTS THEREOF;
MECHANICAL (INCLUDING ELECTROMECHANICAL) TRAFFIC SIGNALLING
EQUIPMENT OF ALL KINDS**

8602 Other rail locomotives; locomotive tenders

8602 10 00 Diesel-electric locomotives

Most diesel engines used for traction are diesel-electric engines.

8603 Self-propelled railway or tramway coaches, vans and trucks, other than those of heading 8604

8603 10 00 Powered from an external source of electricity

See the HS Explanatory Note to heading 8603, third paragraph, (A).

8606 Railway or tramway goods vans and wagons, not self-propelled

8606 91 10 Specially designed for the transport of highly radioactive materials (*Euratom*)

To be classified in this subheading, the railway and tramway goods vans, goods wagons and trucks must incorporate, as an integral part, protective shielding which ensures effective protection against radiation.

8606 91 80 Other

This subheading includes insulated or refrigerated vans and wagons, other than those of subheading 8606 10 00.

Insulated vans and wagons are vehicles which have a cooling source (ice, dry ice, eutectic plates, liquefied gas, etc.) other than a refrigerating device.

Refrigerated vans and wagons are insulated vehicles equipped with a refrigerating device (functioning by compression, absorption or some other means).

8607 Parts of railway or tramway locomotives or rolling stock

8607 11 00 to 8607 19 90 Bogies, bissel-bogies, axles and wheels, and parts thereof

These subheadings include:

1. bogies and bissel-bogies for locomotives;
2. bissel-bogies which are a combination of a bogie and a bissel and which are also used for locomotives;
3. motor bogies (bogies with a built-in electric motor) for rail-cars, railway coaches, vans, trucks and locomotives;
4. bogies for coaches and goods vans.

These subheadings also include parts of bogies and bissel-bogies, such as hydraulic shock absorbers for bogies.

However, springs are excluded from these subheadings (heading 7320).

These subheadings also include the axles, assembled or not, and the wheels and parts thereof referred to in the HS Explanatory Note to heading 8607, second paragraph, (2) and (3).

Rubber tyres and tyre-cases are not classified as parts of wheels in these subheadings (heading 4011 or 4012, as appropriate).

CHAPTER 87

VEHICLES OTHER THAN RAILWAY OR TRAMWAY ROLLING STOCK, AND PARTS
AND ACCESSORIES THEREOF

General

1. Throughout the Combined Nomenclature, 'new vehicles' means vehicles which have never been registered.
2. Throughout the Combined Nomenclature, 'used vehicles' means vehicles which have been registered at least once.

8701 Tractors (other than tractors of heading 8709)**8701 10 00 Single axle tractors**

This subheading covers the items of equipment described in the HS Explanatory Note to heading 8701, sixth and seventh paragraphs, including tracked pedestrian-controlled tractors. Tractors of this type are used mainly in horticulture.

Interchangeable implements intended for use with pedestrian-controlled tractors (harrows, ploughs, etc.) are still classified separately, even if they are mounted on a pedestrian-controlled tractor.

However, if the implements or tools are permanently attached to a chassis with an engine and, together with the latter, form a single mechanical unit, that unit is classified in the heading which covers the working unit. This applies to motorised ploughs and motorised cultivators (heading 8432).

8701 30 00 Track-laying tractors

This subheading includes tractor-type vehicles with extra wide tracks for levelling and smoothing the snow on ski slopes etc.

Machines and working tools designed for fitting to this type of vehicle, as interchangeable equipment (pusher-blades, rotary snow-ploughs, etc.) remain classified in their respective headings (headings 8430, 8479, etc.) even if presented with the vehicle, and whether or not mounted on it.

8701 91 10 to 8701 95 90 Other, of an engine power

These subheadings include so-called 'all-terrain vehicles', designed to be used as tractors, with the following characteristics:

- a handlebar with two grips incorporating the controls for steering the vehicle; the steering is achieved by turning the two front wheels and is based on a motor-car-type steering system (Ackerman principle),
- brakes on all wheels,
- an automatic clutch and a reverse gear,
- an engine specially designed for use in difficult terrain and capable in low ratio of delivering sufficient power to tow attached equipment,
- power transmission to the wheels by shafts and not with a chain,
- tyres having a deep tread design suitable for rough terrain,
- a coupling device of any kind, for example a towing hitch, designed to enable the vehicle to pull or push at least twice its dry weight,
- a towing capacity of a non-braked trailer of twice its dry weight or more. This can be proved by technical documentation, user manual, certificate from the manufacturer or from a national authority, specifying exactly in kilograms the towing capacity of the all-terrain vehicle and its dry weight (the weight of the vehicle without any liquids, passengers or cargo).

If they meet all of the above characteristics and are in accordance with the explanatory note to subheading 8701 91 10, the vehicles are to be classified as agricultural or forestry tractors. Otherwise, they fall under subheadings 8701 91 90, 8701 92 90, 8701 93 90, 8701 94 90 and 8701 95 90.

If they do not meet the above characteristics, the so-called 'all-terrain vehicles' are to be classified in heading 8703.

These subheadings also exclude so-called 'Quads' (heading 8703 or subheading 9503 00 10 (see the explanatory note to that subheading)).

8701 91 10**Agricultural tractors and forestry tractors, wheeled**

These subheadings cover agricultural or forestry tractors having at least three wheels and obviously intended, given their construction and equipment, to be used for agricultural, horticultural or forestry purposes.

Vehicles of this type normally have a maximum speed of 45 km/h.

Their engines are capable of delivering maximum pulling force, for example when using a locking differential.

Their tyres have a deep tread design suitable for agricultural, horticultural or forestry purposes.

Agricultural tractors are generally equipped with a hydraulic device enabling agricultural machinery (harrows, ploughs, etc.) to be raised or lowered, a power take-off enabling the power from the engine to be used to operate other machines or implements, and a coupling device for trailers. They may also be fitted with a hydraulic device intended to operate handling equipment (hay loaders, manure loaders, etc.) when these may be considered to be accessories.

These subheadings also include specially built agricultural tractors, such as raised-chassis tractors (straddle tractors) used in vineyards and nurseries, and hill tractors and tool-carrying tractors.

Even if mounted on the tractor at the time of presentation, interchangeable agricultural equipment remains classified in its respective heading (heading 8432, 8433, etc.).

A particular feature of forestry tractors is the presence of a permanently attached winch enabling timber to be hauled.

In accordance with note 2 to this chapter, tractors falling in these subheadings may also incorporate certain characteristics enabling them, in line with their main purpose, to carry agricultural or forestry machinery, tools, fertilisers, seeds, etc.

These subheadings do not include lawnmowers (referred to as ride-on lawnmowers or garden tractors), fitted with a permanent cutting device and no more than one power take-off solely for the purpose of driving the cutting equipment (see the explanatory note to subheadings 8433 11 10 to 8433 19 90).

**8701 92 10
and
8701 93 10****Agricultural tractors and forestry tractors, wheeled**

See the explanatory note to subheading 8701 91 10.

**8701 94 10
and
8701 95 10****Agricultural tractors and forestry tractors, wheeled**

See the explanatory note to subheading 8701 91 10.

8701 91 90**Other**

This subheading includes:

1. public works tractors;
2. single-axled tractors for articulated motor vehicles.

**8701 92 90
and
8701 93 90****Other**

See the explanatory note to subheading 8701 91 90.

**8701 94 90
and
8701 95 90****Other**

See the explanatory note to subheading 8701 91 90.

8703**Motor cars and other motor vehicles principally designed for the transport of persons (other than those of heading 8702), including station wagons and racing cars**

This heading includes 'multipurpose vehicles', such as motor vehicles which can transport both persons and goods:

1. Of the pick-up type:

This type of vehicle usually has more than one row of seats and is formed by two separate areas, namely a closed cabin for the transport of persons and an open or covered area for the transport of goods.

However, such vehicles are to be classified in heading 8704, if the maximum inner length at the floor of the area for the transport of the goods is more than 50 % of the length of the wheelbase of the vehicle or if they have more than two axles.

2. Of the van-type:

The van-type vehicle with more than one row of seats must fulfil the indications given in the HS Explanatory Note to heading 8703.

However, van-type vehicles with one row of seats, and no presence of permanent anchor points and fittings for installing seats and safety equipment in the rear part of the vehicle, whether or not they have a permanent panel or barrier between the area for persons and the load area or windows in the side panels, are to be classified in heading 8704.

8703 21 10
to
8703 24 90

Other vehicles, with only spark-ignition internal combustion reciprocating piston engine

For the definition of cylinder capacity, see the HS Explanatory Note to subheadings 8407 31, 8407 32, 8407 33 and 8407 34.

These subheadings include 'station wagons' and 'multipurpose vehicles' referred to in the HS Explanatory Note to heading 8703, third and fourth paragraphs.

These subheadings also include small racing cars (called 'skelters' or 'go-carts') without a body, fitted with a spark-ignition internal combustion reciprocating piston engine and capable of reaching relatively high speeds.

8703 31 10
to
8703 33 90

Other vehicles, with only compression-ignition internal combustion piston engine (diesel or semi-diesel)

These subheadings include 'station wagons' and 'multipurpose vehicles' referred to in the HS Explanatory Note to heading 8703, third and fourth paragraphs.

8704

Motor vehicles for the transport of goods

The explanatory note to heading 8703 applies, *mutatis mutandis*.

For the definition of cylinder capacity, see the HS Explanatory Note to subheadings 8407 31, 8407 32, 8407 33 and 8407 34.

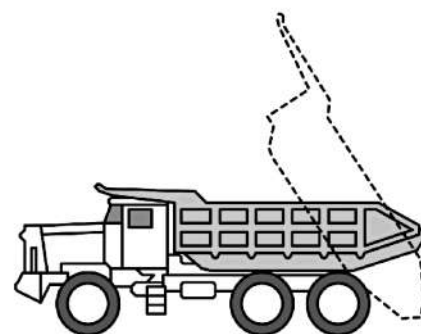
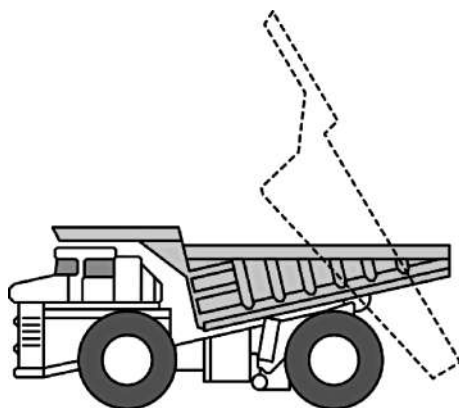
This heading includes four-wheel drive, articulated chassis, all-terrain vehicles in which the front section houses a diesel engine and a cab fitted with controls. The rear section consists of a two-wheeled chassis, without equipment, but designed to be fitted with a variety of equipment.

This heading does not, however, include such vehicles when fitted with agricultural or other special purpose equipment (heading 8705).

8704 10 10
and
8704 10 90

Dumpers designed for off-highway use

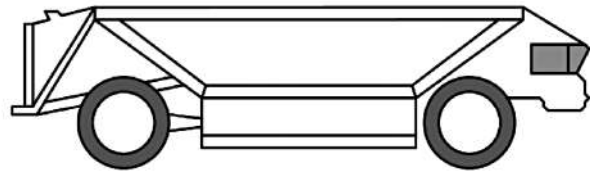
1. These subheadings mainly cover vehicles fitted with a front or rear tipping body or a bottom-opening body that have been specially designed to transport sand, gravel, earth, stones, etc., and are intended for use in quarries, mines, or on building sites, at roadworks, airports and ports. Examples illustrating various types of dumper are given at the end of this note.
2. These subheadings also cover smaller vehicles of the type used on construction sites for carrying earth, rubble, fresh cement and concrete, etc. These have a fixed or articulated chassis and two-or four-wheel drive, the dumper hopper being located above one axle and the driver's seat above the other. The driver's seat is not usually inside a cab.



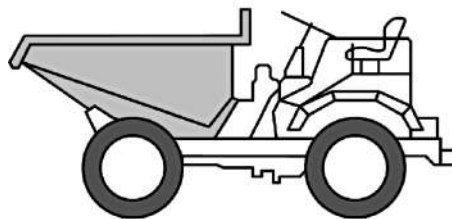
Typical dumpers



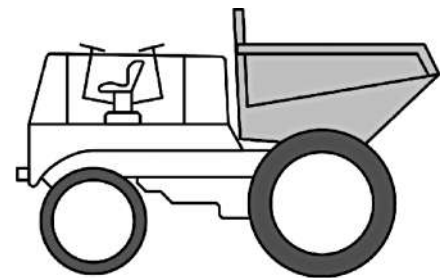
Dumper with two-way seat and two sets of controls



Dumper with bottom-opening body



Dumper for construction sites



Dumper with two-way seat and two sets of controls

8704 21 10

Specially designed for the transport of highly radioactive materials (Euratom)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8704 21 31
to
8704 21 99

Other

These subheadings include 'multipurpose vehicles' referred to in the HS Explanatory Note to heading 8704, fourth paragraph.

8704 22 10

Specially designed for the transport of highly radioactive materials (Euratom)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8704 23 10

Specially designed for the transport of highly radioactive materials (Euratom)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8704 31 10

Specially designed for the transport of highly radioactive materials (Euratom)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8704 31 31
to
8704 31 99

Other

These subheadings include the 'multipurpose vehicles' referred to in the HS Explanatory Notes to heading 8704, fourth paragraph.

8704 32 10

Specially designed for the transport of highly radioactive materials (Euratom)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8707 Bodies (including cabs), for the motor vehicles of headings 8701 to 8705**8707 10 10 For industrial assembly purposes**

For the purposes of this subheading, the expression 'for industrial assembly' applies only to bodies used in a motor-vehicle assembly or manufacturing plant (or bodies used by subcontractors in subassemblies) for the production of new vehicles.

This subheading can be applied only to bodies actually used in the production of new vehicles covered by the subheading itself. Consequently, it does not apply to similar bodies for use as spare parts.

8707 90 10 For the industrial assembly of: Pedestrian-controlled tractors of subheading 8701 10; Vehicles of heading 8704 with either a compression-ignition internal combustion piston engine (diesel or semi-diesel) of a cylinder capacity not exceeding 2 500 cm³ or with spark-ignition internal combustion piston engine of a cylinder capacity not exceeding 2 800 cm³; Special purpose motor vehicles of heading 8705

See the explanatory note to subheading 8707 10 10.

For the definition of 'cylinder capacity', see the HS Explanatory Note to subheadings 8407 31, 8407 32, 8407 33 and 8407 34.

8708 Parts and accessories of the motor vehicles of headings 8701 to 8705

The explanatory note to subheading 8707 10 10 applies, *mutatis mutandis*, to parts and accessories intended for industrial assembly.

8708 70 91 Wheel centres in star form, cast in one piece, of iron or steel

The wheel centres of this subheading are generally used on motor buses or on vehicles for the transport of goods. They are star-shaped, usually with five or six branches, and are designed for use with detachable rims.

8708 70 99 Other

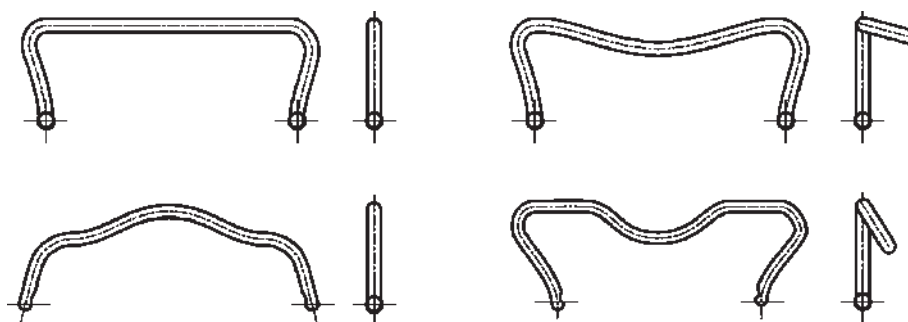
In addition to the parts and accessories referred to in the HS Explanatory Note to heading 8708, second paragraph, (K), this subheading includes wheel-balance weights.

8708 80 55 Anti-roll bars; other torsion bars

Anti-roll bars are springs for vehicles for transmitting suspension forces from one side of the vehicle to the other.

They consist mostly of steel bars of circular cross-section, generally bent to form an approximate U shape.

They typically have the following appearance:



Torsion bars are steel bars which are generally round, but may sometimes be square or bundles of a number of rectangular bars.

Torsion bars are linear, i.e. the torque applied at one end of the bar is proportional to the angle of torsion it produces. They typically have the following appearance:



**8708 99 10
to
8708 99 97**

Other

These subheadings do not include:

- (a) chassis of motor vehicles of headings 8702 to 8704, not fitted with an engine, but fitted with a cab (heading 8702, 8703 or 8704);
- (b) headrests for motor car seats (heading 9401 or 9404).

8709

Works trucks, self-propelled, not fitted with lifting or handling equipment, of the type used in factories, warehouses, dock areas or airports for short-distance transport of goods; tractors of the type used on railway station platforms; parts of the foregoing vehicles

8709 11 10

Specially designed for the transport of highly radioactive materials (*Euratom*)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8709 19 10

Specially designed for the transport of highly radioactive materials (*Euratom*)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

8712 00

Bicycles and other cycles (including delivery tricycles), not motorised

This heading includes incomplete bicycles that have the essential characteristics of complete bicycles (general rule 2(a) for the interpretation of the Combined Nomenclature).

An incomplete bicycle, whether or not assembled, is to be classified under heading 8712 00 if it consists of the frame, the fork and at least two of the following components:

- a set of wheels;
- a crank-gear (see the explanatory note to subheading 8714 96 30);
- a steering unit (including handlebar and handlebar stems);
- a brake system.

8713

Carriages for disabled persons, whether or not motorised or otherwise mechanically propelled

8713 90 00

Other

Motorised vehicles specifically designed for disabled persons are distinguishable from vehicles of heading 8703, mainly because they have:

- a maximum speed of 10 km per hour, i.e. a fast walking pace;
- a maximum width of 80 cm;
- two sets of wheels touching the ground;
- special features to alleviate the disability (for example, footrests for stabilising the legs).

Such vehicles may have:

- an additional set of wheels (anti-tips);
- steering and other controls (for example, a joystick) that are easy to manipulate; such controls are usually attached to one of the armrests; they are never in the form of a separate, adjustable steering column.

This subheading includes electrically-driven vehicles similar to wheelchairs which are only for the transport of disabled people. They can have the following appearance:



However, motor-driven scooters (mobility scooters) fitted with a separate, adjustable steering column are excluded from this subheading. They can have the following appearance and are classified in heading 8703:



8714

Parts and accessories of vehicles of headings 8711 to 8713

8714 91 10 to 8714 99 90

Other

These subheadings include parts and accessories intended for the manufacture, equipping or repair of:

1. side-cars for motorcycles and bicycles;
2. cycles fitted with an auxiliary motor, i.e., cycles which may be propelled by means of pedals and which are fitted with an auxiliary motor (having a cylinder capacity not exceeding 50 cm³);
3. other cycles (including delivery tricycles) not fitted with a motor.

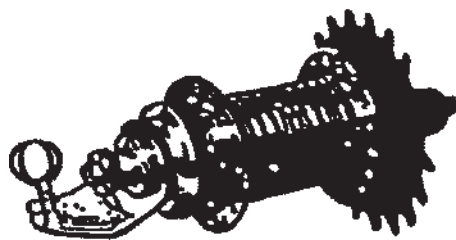
8714 94 20**Brakes**

This subheading includes coaster braking hubs and hub brakes.

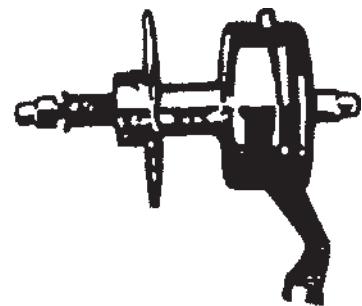
Braking hubs are generally of the back-pedal type.

However, in drum hub brakes, the braking action is produced manually, via a cable or a rod.

They typically have the following appearance:



Braking hub of the back-pedal type



Drum hub brake

8714 94 90**Parts**

This subheading includes brake levers.

However, this subheading does not include rubber brake blocks (subheading 4016 99 97) or brake guides (generally, subheading 8307 10 00 or 8307 90 00).

8714 96 30**Crank-gear**

A crank gear normally consists of:

- a bracket-bearing axle (see image 1),
- one or several gearings (also referred to as sprockets, chain-rings or chain-wheels) usually attached to the right crank/pedal arm (see image 2), and
- a left crank/pedal arm (see image 3).

By application of general rule 2(a) for the interpretation of the Combined Nomenclature, an article remains classified as an incomplete article, whether or not assembled, under this subheading, for example, when it consists of:

- one or several gearings usually attached to the right crank/pedal arm,
- one or several gearings usually attached to the right crank/pedal arm and the left crank/pedal arm, or
- one or several gearings usually attached to the right crank/pedal arm and the bracket-bearing axle.

Image 1



Image 2



Image 3

**8714 99 50****Dérailleur gears**

Dérailleur gears of this subheading are mechanisms that enable the chain of a bicycle to be moved on the front sprockets or on the rear sprockets. A complete dérailleur gear set usually consists of at least one dérailleur (front or rear, see images) and a shift lever attached to the dérailleur by a cable. When a rider operates the lever while pedalling, the change in cable tension moves the dérailleur from side to side, 'derailing' the chain onto different sprockets.

This subheading also covers front or rear dérailleurs presented separately, as well as sets consisting of these dérailleurs with cables or shift levers unassembled. However, presented separately, shift levers and cables are to be classified in subheading 8714 99 90 as parts.



Front dérailleur



Rear derailleur

8714 99 90

Other; parts

This subheading includes child seats intended for the transport of children on 'adult bicycles'. They can be either mounted on the luggage carrier, the frame or can be fitted on or to the handlebars. These seats are suitable for use principally with bicycles and are therefore considered as an accessory for a bicycle.

8716

Trailers and semi-trailers; other vehicles, not mechanically propelled; parts thereof

**8716 10 92
and
8716 10 98**

Trailers and semi-trailers of the caravan type, for housing or camping

In the descriptions, the term 'weight' is to be taken to mean the weight of the vehicle with all of the permanent fixtures and fittings.

8716 39 10

Specially designed for the transport of highly radioactive materials (*Euratom*)

The explanatory note to subheading 8606 91 10 applies, *mutatis mutandis*.

CHAPTER 88

AIRCRAFT, SPACECRAFT, AND PARTS THEREOF**Subheading note 1**

Permanently fitted items of equipment do not include, for example, emergency equipment (for example, lifeboats, parachutes and escape chutes) and interchangeable equipment for weapons systems.

Where an incomplete or unfinished machine is classified as a complete article under general rule 2(a) for the interpretation of the Combined Nomenclature, the normal operating weight is to be used for determining the appropriate subheading.

8802

Other aircraft (for example, helicopters, aeroplanes); spacecraft (including satellites) and suborbital and spacecraft launch vehicles

**8802 11 00
and
8802 12 00****Helicopters**

These subheadings cover only machines in which lift and propulsion are achieved by means of one or more engine-driven rotors.

8802 11 00**Of an unladen weight not exceeding 2 000 kg**

This subheading covers multi-rotor helicopters (so-called 'drones') of different sizes which can operate autonomously (to a pre-programmed destination) or are remotely piloted by the user.

However, if they have the character of toys or recreational models, they are classified under heading 9503. See the Explanatory notes to the Combined Nomenclature to that heading.

SECTION XVIII

**OPTICAL, PHOTOGRAPHIC, CINEMATOGRAPHIC, MEASURING, CHECKING,
PRECISION, MEDICAL OR SURGICAL INSTRUMENTS AND APPARATUS; CLOCKS
AND WATCHES; MUSICAL INSTRUMENTS; PARTS AND ACCESSORIES THEREOF**

CHAPTER 90

**OPTICAL, PHOTOGRAPHIC, CINEMATOGRAPHIC, MEASURING, CHECKING,
PRECISION, MEDICAL OR SURGICAL INSTRUMENTS AND APPARATUS; PARTS AND
ACCESSORIES THEREOF****9001**

Optical fibres and optical-fibre bundles; optical-fibre cables other than those of heading 8544; sheets and plates of polarising material; lenses (including contact lenses), prisms, mirrors and other optical elements, of any material, unmounted, other than such elements of glass not optically worked

This heading includes articles used with visible light and articles used with the invisible part of the spectrum (infra-red and ultraviolet).

The heading does not include, however, electronic 'optical' elements, for example, electrostatic lenses, electromagnetic lenses and so-called field lenses (generally, Chapter 85).

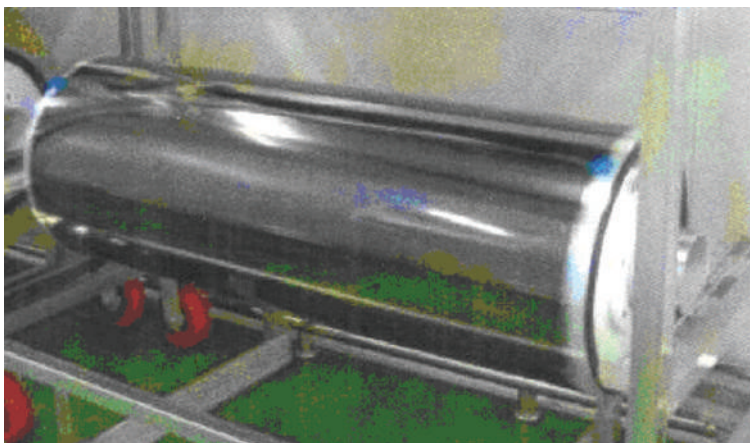
**9001 10 10
and
9001 10 90****Optical fibres, optical-fibre bundles and cables**

These subheadings do not include connectors and adapters for connecting optical fibres, optical-fibre bundles and optical-fibre cables.

See also the explanatory note to subheading 8536 70 00.

9001 20 00**Sheets and plates of polarising material**

This subheading includes foils of polarising material in rolls, for example, for use in the manufacture of LCD modules.

**9001 90 00****Other**

This subheading includes:

1. rubies and other optical elements for lasers;
2. Fresnel lenses of plastics, intended for use, for example, after fitting with a mounting, as magnifying screens for television receivers.

9004	Spectacles, goggles and the like, corrective, protective or other Cords, chains and the like (for example, 'spectacle cords' or 'spectacle chains') for the products of this heading, whether or not with end loops, are to be considered as neither parts nor accessories, because they are neither integral components nor do they add to or improve the functionality of these products. Therefore, they are to be classified according to their constituent material (for example, heading 5609, 6307, 7117, 7315 or 7616).
9005	Binoculars, monoculars, other optical telescopes, and mountings therefor; other astronomical instruments and mountings therefor, but not including instruments for radio astronomy This heading includes apparatus using image intensifiers for night vision.
9006	Photographic (other than cinematographic) cameras; photographic flashlight apparatus and flashbulbs other than discharge lamps of heading 8539
9006 59 00	Other This subheading covers cameras of a kind used for preparing printing plates or cylinders. See the HS Explanatory Note to heading 9006, part (I), third paragraph, (17).
9010	Apparatus and equipment for photographic (including cinematographic) laboratories, not specified or included elsewhere in this chapter; negatoscopes; projection screens
9010 50 00	Other apparatus and equipment for photographic (including cinematographic) laboratories; negatoscopes This subheading includes printed circuit board exposure apparatus which copies photographic negatives of the circuit patterns of printed circuits by exposure onto boards of insulating material from which printed circuit boards are made. The apparatus basically comprises an exposure chamber fitted with ultraviolet lamps, into which the photographic negative and the board of insulating material are placed and in which the board is exposed in contact with the negative in a vacuum.
9013	Liquid crystal devices not constituting articles provided for more specifically in other headings; lasers, other than laser diodes; other optical appliances and instruments, not specified or included elsewhere in this chapter
9013 80 90	Other This subheading includes magnifying screens for television sets, consisting of an optical element (Fresnel lens) of plastics, a frame and a system of metal rods specially designed to fix the screen in front of the television set.
9017	Drawing, marking-out or mathematical calculating instruments (for example, drafting machines, pantographs, protractors, drawing sets, slide rules, disc calculators); instruments for measuring length, for use in the hand (for example, measuring rods and tapes, micrometers, callipers), not specified or included elsewhere in this chapter
9017 10 90	Other This subheading includes drafting tables equipped with pantographs or the like.
9017 20 05 to 9017 20 90	Other drawing, marking-out or mathematical calculating instruments These subheadings include: 1. coordinatographs not designed for photogrammetry; 2. lettering stencils clearly identifiable as being specialised as drawing or marking-out instruments.

9018 Instruments and appliances used in medical, surgical, dental or veterinary sciences, including scintigraphic apparatus, other electromedical apparatus and sight-testing instruments

9018 50 10 Non-optical

In addition to ultrasonic diagnostic equipment for general use, this subheading includes special appliances for ultrasonic examination of the eyes (such as those for determining the thickness of the cornea and eye lens or the length of the eyeball).

9018 90 84 Other

This subheading includes:

1. electrical defibrillators for delivering current pulses to restore the natural heart function. With these instruments, which are equipped with a pulse generator and two defibrillation electrodes, the patient's electrocardiogram (ECG) signal coming from the electrodes is displayed on a screen or printed out on a built-in printer;
2. medical appliances for blowing gases into the human abdominal cavity in order to make it possible to examine various organs with an endoscope. Attached to these appliances, which are provided with measuring instruments and displays, are two hoses which are connected at their ends to a stopcock and a long needle;
3. medical suction pumps for pumping out secretions, comprising the pump and a suction device and used in operating theatres or in ambulances;
4. contraceptive appliances known as 'intra-uterine devices (IUDs)', made of plastics together with copper wire, copper in colloidal form or a hormone insert.

This subheading does not cover so-called 'tourniquets'. These generally consist of a strap and a buckle system. They are used, by tightening the strap, to control the blood circulation to an extremity for a short period. Other kinds of tourniquets may have, for example, a rod for tightening. Due to their simple construction and materials used, they are not similar to instruments and appliances covered by the present subheading although they fulfill a medical purpose (generally, classified according to the constituent material of the strap). See in particular the HS Explanatory Note to heading 9018, fourth paragraph.



9021 Orthopaedic appliances, including crutches, surgical belts and trusses; splints and other fracture appliances; artificial parts of the body; hearing aids and other appliances which are worn or carried, or implanted in the body, to compensate for a defect or disability

For the purposes of this heading, the expression 'to compensate for a defect or disability' means only appliances which actually take over or substitute for the function of the defective or disabled part of the body.

This heading does not include appliances, which simply alleviate the effects of the defect or disability.

This heading also excludes appliances identifiable for ostomy use (subheading 3006 91 00).

9021 10 10**Orthopaedic appliances**

This subheading includes 'orthopaedic appliances' which are specially designed for a particular orthopaedic purpose, in distinction to ordinary products that might be used for different purposes (for example, products for overstrained joints, ligaments or tendons caused by sporting activities, type writing, and products which simply alleviate pain in the defective or disabled part of the body, for example, caused by inflammation).

The 'orthopaedic appliances' must completely prevent a specific movement of the defective or disabled part of the body (for example, joints, ligaments or tendons) in order to exclude further injuries or (an aggravation of) bodily deformities as distinct from ordinary products that allow undesired movements, yet avoid reflex movements (i.e. movements carried out subconsciously) through their relative inflexibility caused by, for example, flexible splints, pressure pads, non-elastic textile materials, restrictions through 'Velcro-type' straps.

See also Note 6 to Chapter 90 and the judgment in Joined Cases C-260/00 to C-263/00, *Lohmann GmbH & Co. KG and Medi Bayreuth Weihermüller & Voigtmann GmbH & Co. KG v Oberfinanzdirektion Koblenz*, paragraphs 36, 37, 39, 40, 43 and 45, (ECLI:EU:C:2002:637).

Examples of products falling under subheading 9021 10 10:



Examples of products falling under Section XI:



9021 39 90

Other

This subheading includes:

1. plates which remain permanently in the body, for example, to replace part or the whole of a bone);
2. woven strips of man-made fibre, which are implanted in the knee joint to replace the ligaments in the case of chronic knee-joint ligament instabilities.

9021 40 00

Hearing aids, excluding parts and accessories

This subheading includes the appliances covered by the HS Explanatory Note to heading 9021, part (IV), even if in the form of spectacles.

9021 50 00

Pacemakers for stimulating heart muscles, excluding parts and accessories

This subheading includes only cardiac pacemakers proper. Subject to notes 1 and 2 to this chapter, parts and accessories for such pacemakers, such as casings, casing halves, casing covers and electrodes fall in subheading 9021 90 90.

Separately presented primary batteries and electric accumulators for pacemakers are classified in heading 8506 or 8507. Chargers with a transformer primary for charging the accumulator in the implanted pacemaker by inducing a current in the secondary in the pacemaker fall in heading 8504.

9021 90 90

Other

This subheading includes the following appliances which compensate for a defect or a disability:

1. appliances implanted in the human body as medicament dispensers and accommodating in a common housing a 'medicinal pump', the power source for the pump and the reservoir for the medicament;
2. 'annular prostheses', i.e., stainless steel rings covered with two layers of plastics and a man-made fibre fabric. These are surgically attached to the heart valve in order to restore the heart valve's ability to close, for example, in the case of mitral insufficiency;

3. 'umbrella' filters for implanting in the cardiac vein (vena cava inferior) in order to prevent thromboses and embolisms from migrating to the heart. They consist of a very small, umbrella-shaped frame made of a stainless steel alloy, which is covered with a thin layer of synthetic elastic material and opens like an umbrella in the vein;
4. permanent ureter dilators of plastics. These barbed, rod-shaped appliances are inserted into the ureter to permit urine to flow.

9022 **Apparatus based on the use of X-rays or of alpha, beta or gamma radiation, whether or not for medical, surgical, dental or veterinary uses, including radiography or radiotherapy apparatus, X-ray tubes and other X-ray generators, high-tension generators, control panels and desks, screens, examination or treatment tables, chairs and the like**

9022 12 00 **Computed tomography apparatus**

See the HS Explanatory Note to subheading 9022 12.

However, video recording systems, which are not incorporated in X-ray apparatus and which digitise, edit and store analogue video signals supplied by an external video camera fall in heading 8543. They essentially comprise an analogue-to-digital converter, a process computer, monitors and a magnetic tape or disk memory.

9022 90 20 **Parts and accessories of apparatus based on the use of X-rays**

This subheading includes beryllium windows for X-ray tubes.

9025 **Hydrometers and similar floating instruments, thermometers, pyrometers, barometers, hygrometers and psychrometers, recording or not, and any combination of these instruments**

9025 11 20 **Liquid-filled, for direct reading**

and
9025 11 80

Thermometers 'for direct reading' are of a kind in which the temperature is indicated on a scale by the level of the fluid in the thermometer.

9026 **Instruments and apparatus for measuring or checking the flow, level, pressure or other variables of liquids or gases (for example, flow meters, level gauges, manometers, heat meters), excluding instruments and apparatus of heading 9014, 9015, 9028 or 9032**

9026 20 20 **For measuring or checking pressure**

to
9026 20 80

These subheadings include tyre inflators incorporating a pressure gauge and are regarded as pressure gauges even if they are not designed to be connected to an external supply source but incorporate their own compressed-air reservoir.

9027 **Instruments and apparatus for physical or chemical analysis (for example, polarimeters, refractometers, spectrometers, gas or smoke analysis apparatus); instruments and apparatus for measuring or checking viscosity, porosity, expansion, surface tension or the like; instruments and apparatus for measuring or checking quantities of heat, sound or light (including exposure meters); microtomes**

9027 10 10 **Electronic**

This subheading includes laser air-particle counters. These are electronic devices which determine or monitor the dust content of filtered air, for example, in industrial plants or in the medical field. The dust particles contained in the air sample cause a laser beam to produce scattered light in the measurement chamber of the device which, after being concentrated into a beam by a lens system, is picked up by a photodiode and converted into an electrical signal. With the aid of pre-programmed reference data, the proportion of dust particles is determined and the measurement result is shown on the instrument's digital display or printed out on an external tape printer. Via an interface, the measurement result in the form of an electrical signal can also be sent by cable to an automatic data-processing machine.

9027 30 00 Spectrometers, spectrophotometers and spectrographs using optical radiation (UV, visible, IR)

This subheading includes microprocessor-controlled electronic devices (multichannel optical analysers) for measuring and analysing the wavelengths of optical signals for spectral analyses. The wavelengths of the optical signals, measured by detectors, are converted into digital electrical signals and compared with reference values (analysed). The result of the comparison is evaluated by computer and displayed on external monitors which can be connected up to the equipment.

9027 50 00 Other instruments and apparatus using optical radiation (UV, visible, IR)

This subheading includes electronic devices used in hospital chemical laboratories for the fully automatic analysis of blood sera. They consist essentially of the analyser proper (with sample preparation device, reagent metering mechanism and photometric measurement system using a halogen lamp as the light source and photodiodes as the detectors), a control and evaluation device (with microprocessors and screen for displaying the measurement results) and a printer for recording the results. All three devices are interconnected by electric cables.

9027 80 99 Other

This subheading includes climatic testing cupboards equipped with a pressure chamber, an electric heater, an air humidifier and an electric unit, in which electronic components are exposed to specific pressure, temperature and humidity conditions, simulating the influences and environmental conditions occurring in their subsequent use, in order to test their service life, insulation, etc.

9030 Oscilloscopes, spectrum analysers and other instruments and apparatus for measuring or checking electrical quantities, excluding meters of heading 9028; instruments and apparatus for measuring or detecting alpha, beta, gamma, X-ray, cosmic or other ionising radiation

In accordance with general rule 3(b) for the interpretation of the Combined Nomenclature, instruments and apparatus which can be used for measuring or checking both electrical and non-electrical quantities still fall in heading 9030 if their nature determines that they are mainly used for measuring or checking electrical quantities. This is the case, for example, with cathode ray oscilloscopes and oscillographs and light-beam and UV-beam oscillographs (subheading 9030 20 00).

However, instruments and apparatus whose essential character cannot be determined because they are used for measuring or checking both electrical and non-electrical quantities to the same extent do not fall in this heading. Accordingly, in accordance with general rule 3(c) for the interpretation of the Combined Nomenclature, instruments for testing motor vehicle engines and ignition systems by measuring electrical quantities such as voltage and resistance and non-electrical quantities such as speed of rotation, contact angle and state of contact breaker points fall in heading 9031.

9030 20 00 Oscilloscopes and oscillographs

This subheading includes light-beam and UV-beam oscillographs for measuring and recording rapid variations of electrical quantities. These instruments, also known as light-beam recorders, UV recorders or loop oscillographs, record the signals produced by the periodic phenomenon under study by means of a light or UV beam on photosensitive paper.

9030 39 00 Other, with a recording device

This subheading includes electrical testing devices or systems which determine the serviceability of printed circuits or other electronic components and display any defects such as short circuits or breaks by measuring or checking electrical quantities like capacitance, inductance, impedance, resistance and voltage.

These devices or systems usually comprise a measuring and checking unit (with input keyboard, program memory and CRT terminal), which carries out the measurements and compares the results with reference values and displays them, a control unit (with automatic data-processing machine or microprocessors), an output printer for printing out the test results and a device for sorting the tested components according to specific actual values and separates out defective ones.

The subheading does not, however, include electrical devices for checking whether the packages of electronic components are hermetically sealed (heading 9031).

9030 82 00 For measuring or checking semiconductor wafers or devices

This subheading covers electrical testing devices or systems which measure or check electrical quantities such as voltage, frequency, etc., to determine the serviceability of wafers, chips and other semiconductor devices and to display any defects such as deviations from previously set values.

The devices or systems usually comprise a measuring and checking unit (with input keyboard, program memory and CRT terminal), which carries out the measurement and compares the results with reference values and displays them, a control unit (with automatic data-processing machine or microprocessors), an output printer for printing out the test results and a device for sorting the tested components according to specific actual values and separates out defective ones.

The subheading does not, however, include electrical devices for checking whether the packages of integrated circuits or other electronic components are hermetically sealed (heading 9031).

9031 Measuring or checking instruments, appliances and machines, not specified or included elsewhere in this chapter; profile projectors

9031 20 00 Test benches

Test benches for checking fuel-injection pumps of diesel engines have, as essential components which are all fixed to the same stand, an electric motor and a device comprising injectors and graduated glass tubes for checking the output of the injection pump elements; they may additionally be fitted with an auxiliary appliance (a stroboscope) for checking the exact timing of fuel injections.

9031 80 20 For measuring or checking geometrical quantities

Examples of geometrical quantities are: length, distance, diameter, radius, curvature, angle, tilt, volume, and surface roughness.

This subheading also includes spirit levels.

This subheading does not include interferometers for checking plane surfaces, used in laboratories (subheading 9027 50 00).

CHAPTER 91

CLOCKS AND WATCHES AND PARTS THEREOF

9102 Wristwatches, pocket-watches and other watches, including stopwatches, other than those of heading 9101

This heading includes combinations of a watch with an electronic calculating machine which together have the appearance of a wrist-watch or pocket-watch.

It does not include electronic calculators with a time-and-date display and an alarm (subheading 8470 10 00, 8470 21 00 or 8470 29 00).

9111 Watch cases and parts thereof

Watch straps, bands or bracelets attached to watch cases are classified with the watch-cases. However, when presented with watch-cases but not attached thereto, they are classified under their respective heading in the same way as watch straps, bands or bracelets presented separately (heading 9113).

9114 Other clock or watch parts**9114 10 00 Springs, including hairsprings**

This subheading covers all springs used in clock or watch movements.

Examples, other than mainsprings and hairsprings, are:

1. friction springs;
2. controlling springs;
3. ratchet springs, balance springs, tension springs and cross springs.

This subheading does not include springs for clock cases and for cases of a similar type which constitute parts of general use within the meaning of note 2 to section XV.

Mainsprings mounted in their barrels fall in subheading 9114 90 00.

9114 90 00 Other

This subheading includes:

1. assemblies of electric or electronic components forming a recognisable part of a clock — for example, an electronic chime;
2. articles known as 'balance screws' or 'balance cup screws';
3. wedges, generally of plastics, set between the case and the movement of a watch;
4. quartz oscillators for watches (quartz resonators connected with an electronic microcircuit for maintaining oscillations).